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Research article

Dynamic behaviors of keyhole and weld pool during the digging stage in variable polarity plasma arc welding of thick aluminum alloy

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Abstract

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1. Introduction

2. Experimental set-up and procedure

Highlights

- Developed a transparent system for in-situ monitoring of plasma-metal-solid interactions.
- Arc-induced oscillations break thermal-force equilibrium, driving dynamic, multi-phase evolution of the digging stage.
- Correlated observable features with internal keyhole and weld pool dynamics, providing indicators for torch control.

Abstract

Variable polarity plasma arc welding (VPPAW) is a high-energy-density keyhole welding technique with considerable potential for the efficient and defect-free joining of thick aluminum alloy. However, the extended digging depth in thick workpieces intensifies the complexity of the welding process, as torch travel at different digging depths, even under optimized parameter, results in distinct variations in weld quality. In this study, a transparent observation system was established by butt-jointing a thick aluminum alloy plate with a quartz glass sheet, enabling direct visualization of the dynamic behaviors of the keyhole and weld pool during the digging stage. A deep learning-based image processing approach utilizing the SegFormer architecture was developed to extract these dynamic features. The findings reveal that the weld pool depth evolution is a highly dynamic multi-phase process, progressing through three alternating cycles of rapid growth and quasi-steady behavior, ultimately reaching a blasting-type penetration. The mechanism involves the establishment of thermal-force equilibrium when the molten metal reaches a critical thickness, halting the digging process. Arc-induced oscillations break this equilibrium, triggering molten metal outflow and keyhole deepening, which reflects the transient and non-equilibrium characteristics of energy transfer in plasma-metal coupling. Molten metal droplet cluster represents a distinctive phenomenon during the digging stage, arising from the continuous outflow of molten metal from the interior of keyhole. The externally observable weld pool and droplet cluster areas effectively indicate the internal digging stage, offering characteristic features to guide torch travel and ensure weld quality. Observations under varying welding parameters demonstrate that the dynamic, multi-phase digging process persists, confirming its governance by transient, non-equilibrium energy transfer mechanisms. The timing of droplet cluster detachment serves as a crucial indicator for assessing the suitability of welding parameter. This work provides a solid foundation for intelligent control in advanced VPPAW of thick aluminum alloy.

Graphical Abstract

3. Results and discussion

4. Conclusions and future work

CRediT authorship contribution statement

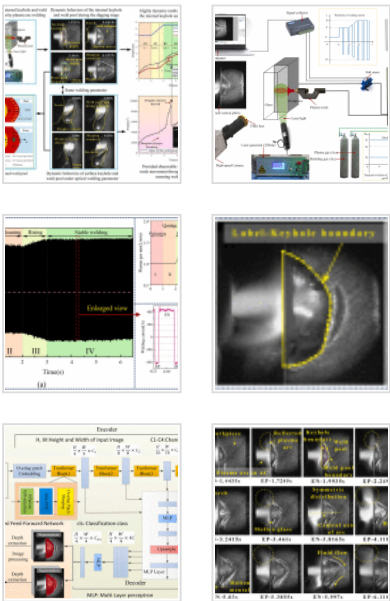
Declaration of Competing Interest

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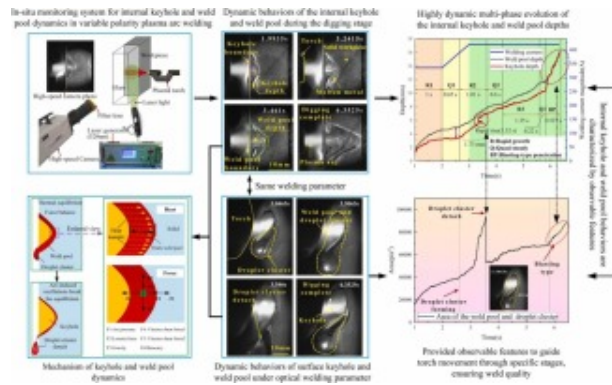
Figures (18)



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Table 1. Typical welding parameters desi...



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Keywords

Variable polarity plasma arc welding; Transparent observation system; Deep learning; Keyhole and weld pool behaviors; Energy transfer

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1. Introduction

Thick aluminum alloys are widely used in industries such as aerospace, transportation, and power equipment manufacturing due to their exceptional strength-to-weight ratio, corrosion resistance, and thermal conductivity [1], [2]. However, welding thick aluminum alloys poses significant challenges due to their high thermal conductivity, strong reflectivity, high melting point oxide layer, and large thermal expansion coefficient [3], [4]. Compared to conventional welding methods for thick aluminum alloys, such as tungsten inert gas welding and metal inert gas welding, variable polarity plasma arc welding (VPPAW) offers distinct advantages, including deeper penetration and higher

welding speed [5], [6]. The alternating polarity in VPPAW provides an effective cathodic cleaning effect that removes the oxide layer on the aluminum surface, improving weld quality and reducing welding defects [7], [8]. During the VPPAW process, keyhole formation is achieved through the high-energy plasma arc melting the material, while the plasma jet force penetrates through the thickness of the workpiece. This enables single-sided welding with double-sided formation, avoiding traditional multi-pass welding and significantly reducing heat distortion and residual stress. Although the keyhole effect is also utilized in laser welding, VPPAW presents a more cost-effective solution with a lower risk of porosity defects [9]. Consequently, VPPAW demonstrates great potential for the welding of thick aluminum alloys.

The VPPAW process for welding aluminum alloys typically comprises three distinct stages: the digging stage (also known as the perforation stage), the initial unstable stage, and the keyhole stable movement stage [10]. Although VPPAW has demonstrated high efficiency and weld quality in aluminum alloys, its application has been predominantly confined to workpieces with thicknesses up to 10 mm. When applied to thicker plate welding, collapses, failures to penetrate, and cutting is prone to occur, significantly influencing weld quality, with these defects particularly concentrated during the initial digging stage [11]. The dynamic evolution of the keyhole and the weld pool serves as a direct manifestation of arc energy distribution and heat transfer mechanisms, thereby exerting a critical influence on welding process stability and weld quality. As a result, considerable attention has been dedicated to elucidating the keyhole dynamic behavior and weld pool evolution during the digging stage, aiming to deepen the understanding of their roles in defect formation and weld performance.

Zhang and Zhang developed a plasma cloud charge sensor to detect keyhole formation during VPPAW, discovering that the plasma cloud signal rapidly diminishes to zero upon the establishment of a fully penetrated keyhole [12]. Zhang et al. proposed an efflux plasma charge sensor to monitor keyhole dynamics, providing insight into the physical phenomena associated with plasma jets and space charge behavior [13]. Song et al. and Saad et al. utilized acoustic emissions under varying welding parameters to identify

different keyhole modes [14], [15]. Although electrical and acoustic signal-based methods can indirectly infer the keyhole state, they generally suffer from low spatial resolution and limited robustness against environmental interference.

To achieve more direct and reliable characterization of keyhole morphology, vision-based techniques have been increasingly adopted. Liu et al. employed a dual CCD camera system to capture the length, width, and aspect ratio of the bottom keyhole, revealing that the exit-side keyhole typically assumes an oblate shape with its center offset rearward from the torch axis [16]. Zhang et al. developed a single-camera vision system capable of simultaneously capturing both the keyhole and the weld pool at the bottom side of the workpiece, and identified six distinct evolutionary stages throughout the welding process. [17]. Liu et al. advanced this approach by simultaneously capturing the topside weld pool and keyhole entrance using dual CCD cameras and integrating the captured images through registration, thereby enabling a holistic characterization of weld pool-keyhole boundaries [18]. Wu et al. demonstrated the weld penetration status (Non penetration, Partial penetration, Full penetration, Cutting) can be revealed by bottom keyhole characteristics [19], [20]. In the subsequent study, they extracted geometric features of the topside keyhole and weld pool to predict the penetration status [21]. Furthermore, Wu et al. improved the accuracy of weld penetration status detection through the fusion of acoustic and bottom keyhole features, which laid a solid foundation of intelligent control for VPPAW [22], [23]. While camera-based observation of the top and bottom keyholes can provide localized surface contour information, it remains challenging to capture the dynamic internal features of the keyhole and weld pool, thereby limiting comprehensive understanding and precise control of the welding process.

The internal dynamic evolution of the keyhole and weld pool can be detailed calculated through numerical simulation techniques. Using a three-dimensional numerical model, Chen et al. demonstrated that during the digging stage in VPPAW of medium-thickness workpieces, the weld pool depth increases rapidly, exhibiting a characteristic "blasting-type" penetration behavior [10]. Pan et al. identified three formation stages associated

flow patterns within the weld pool throughout the digging process by numerical results [24]. In a series of studies, Wei et al. identified the critical conditions leading to keyhole collapse and clarified the mechanisms responsible for defect formation, offering important insights into the digging process [25], [26]. However, due to inherent simplifications in the physical modeling, such as assumptions regarding heat and mass transfer and uncertainties in boundary conditions, the simulation results often deviate from experimental observations. Researchers have employed advanced visual sensing techniques to investigate the characteristics of the keyhole and weld pool. For instance, Miyagi et al. utilized X-ray phase contrast imaging to quantitatively assess the internal geometry of the keyhole [27]. Xu et al. employed in situ X-ray imaging to elucidate material flow behavior and the mechanisms underlying keyhole and weld pool stability [28]. Although X-ray imaging provides direct and intuitive insights into the internal dynamics of the keyhole and weld pool, its widespread adoption is hindered by high equipment costs, operational complexity, and stringent safety requirements.

The "sandwich structure", in which an aluminum alloy is placed adjacent to a high-temperature-resistant glass, has been widely utilized in laser welding to enable in situ visualization of internal keyhole morphology [29], [30]. Recently, Zhou et al. extended this method to direct current (DC) plasma arc welding, capturing through-thickness images of the keyhole and weld pool [11]. This technique facilitated quantitative analysis of keyhole geometry and detailed characterization of weld pool flow dynamics, offering valuable insights for the optimization of keyhole welding. However, its application in VPPAW remains limited due to the periodically reversing current waveform, which causes unsteady arc–molten metal interactions and transient thermal–mechanical fluctuations. In addition, large arc brightness variations during polarity switching interfere with image acquisition, posing significant challenges for efficiently and accurately image processing.

Image processing techniques, such as image thresholding segmentation, morphological operations, and segmented linear stretching, have been widely employed in welding process monitoring to extract the dynamic features of the weld pool and keyhole [31]. For example, Zeng et al. developed keyhole and weld pool image extraction algorithms,

systematically exploring the intrinsic relationship between optical radiation signals and the dynamic behaviors of the keyhole and weld pool [32]. Cai et al. proposed a parabolic mask and binary threshold segmentation based on the morphological characteristics of the weld pool, facilitating the extraction of multiple weld pool and keyhole features from monitoring images [33]. Although these traditional image processing approaches are computationally efficient and straightforward, they are highly susceptible to disturbances such as arc flickering, spatter occlusion, and background reflections, which significantly compromise their robustness and limit their applicability under varying welding conditions [34]. To address these limitations, researchers have increasingly adopted machine learning (ML) techniques to process complex data and improve the robustness of feature extraction under noisy environments. For example, Knaak et al. proposed an ML-based statistical approach to reduce the complexity of the feature space in laser welding by eliminating redundant information [35]. Nalajam et al. employed ML techniques such as K-means clustering, support vector machines, and random forests for porosity segmentation in microstructural images [36]. Despite achieving satisfactory accuracy, these methods often overlooked critical contextual information during image preprocessing, leading to incomplete or inaccurate interpretations of keyhole and weld pool dynamics.

Recently, convolutional neural networks (CNN) have been extensively applied in monitoring keyhole and weld pool states, owing to their ability of automatically extracting latent visual features. By leveraging keyhole and weld pool image datasets, Zhang et al. developed a CNN-based image processing algorithm optimized through systematic tuning of network architecture and hyperparameters to accurately classify different welding penetration states [37]. Cai et al. employed a U-Net architecture built upon CNN for pixel-level segmentation of keyhole and weld pool images, effectively mitigating multiple interference factors and achieving precise delineation of geometric contours [38]. While CNN demonstrate outstanding performance in feature extraction, the inherent pooling operations inevitably lead to the loss of spatial information, making it challenging to preserve fine-grained boundary details of the keyhole and weld pool.

Transformer architectures, which employ self-attention mechanisms, have surpassed CNN across various computer vision tasks, particularly in terms of both accuracy and computational efficiency. Zeng et al. applied a Transformer-based deep learning framework to analyze spatio-temporal features of optical signals from two-dimensional time–frequency diagrams, providing a robust monitoring technology that supports the acceleration of intelligence and automation in laser welding [39]. Furthermore, SegFormer, a lightweight yet highly effective semantic segmentation architecture, combines a Transformer backbone with a compact multilayer perceptron decoder. This design retains the long-range dependency capturing capability of Transformers while enhancing real-time processing efficiency through model simplification [40], [41]. Despite the exceptional adaptability of the SegFormer architecture, its application in plasma arc welding monitoring remains largely unexplored.

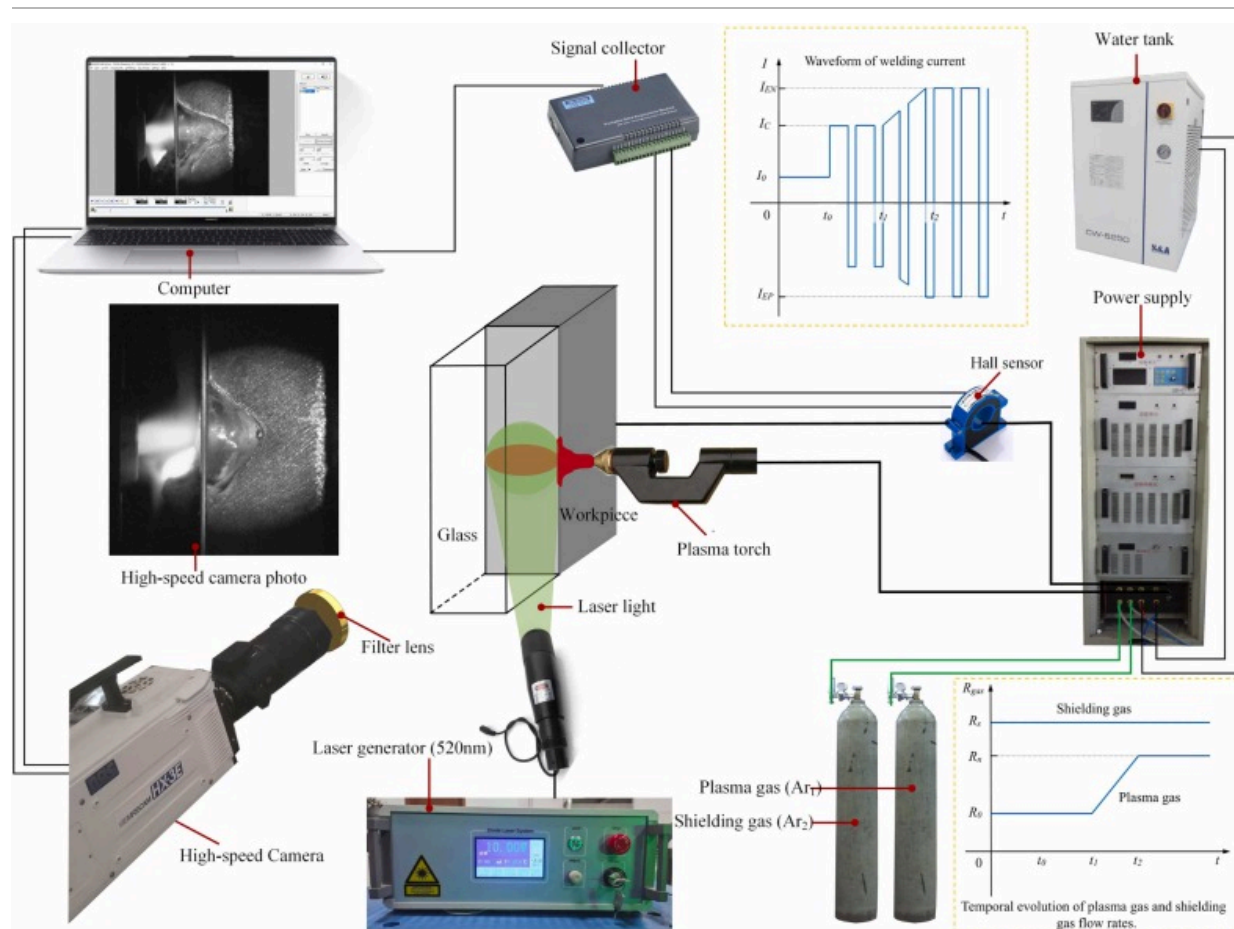
In contrast to medium-thick aluminum alloy plates (3–10 mm), the extended digging distance in thick sections increases the interaction time between the high-energy arc and the material, resulting in an enlarged weld pool volume and intensified convection, thereby complicating the welding process. Even under optimized parameters, torch travel across varying digging depths readily induces defects such as incomplete penetration, undercutting, and cutting, making the achievement of stable weld quality particularly challenging. To address this issue, an optimized sandwich-structured experimental configuration was employed for the first time to elucidate the dynamic evolution of the keyhole and weld pool during the digging stage of VPPAW. A deep learning–based segmentation framework was developed to enable real-time and accurate extraction of keyhole and weld pool features from high-speed image sequences. To meet industrial application demands, synchronous observation of the surface keyhole and weld pool evolution was achieved, establishing correlations between externally observable characteristics and internal dynamics. These correlations provide discriminative features to guide torch motion and ensure weld consistency. Furthermore, the effects of plasma gas flow rate and welding current on keyhole and weld pool behaviors were

systematically investigated. The findings offer practical insights for enhancing process stability and reliability in VPPAW of thick aluminum alloy.

2. Experimental set-up and procedure

2.1. Experimental system

Fig. 1 illustrates the transparent keyhole and weld pool monitoring system based on the “sandwich structure”, which is employed to visualize the dynamic evolution of the keyhole and weld pool during the digging stage in VPPAW. The experimental setup primarily comprises a VPPAW power source, dual argon gas, a cooling water tank, a high-speed camera, a current transformer, and a specialized observation assembly utilizing a quartz glass window. The plasma gas flow rate is adjustable between 1.4 and 1.8 L/min, while the shielding gas flow rate is fixed at 15 L/min. A 16 mm-thick 5052 aluminum alloy plate was selected as the base material, and the welding process was conducted in the vertical-up position. The tungsten electrode setback was set to 4 mm, and the nozzle diameter was 4.8 mm. The arc height was adjusted to 8 mm, deviating from the conventional 4 mm typically employed for medium-thickness plates. This adjustment is particularly necessary due to the formation of a larger weld pool in thick plates, which promotes the outward ejection of molten metal during keyhole formation. The ejected metal easily adheres to the nozzle, potentially leading to welding defects and process instability.



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Fig. 1. Transparent keyhole and weld pool monitoring system during the digging stage in variable polarity plasma arc welding.

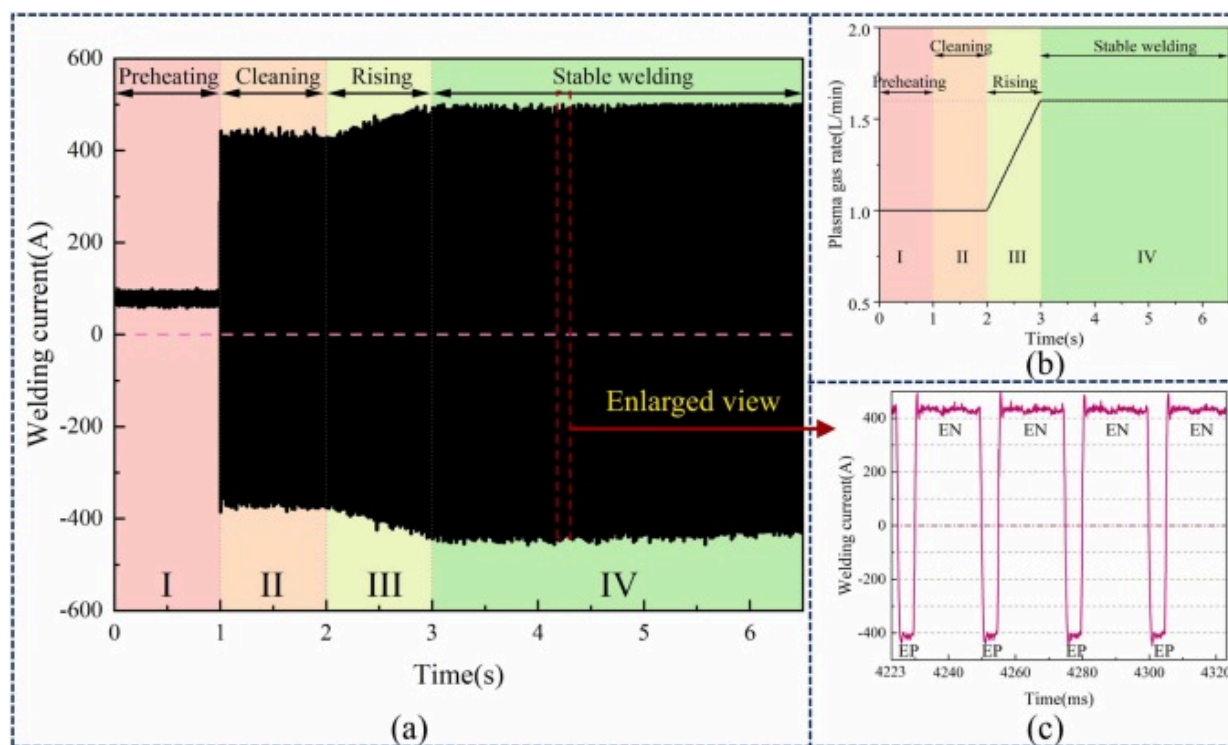
A key feature of this experimental setup is the transparent observation mechanism, which facilitates real-time monitoring of keyhole stability and weld pool dynamics. During the welding process, a high-energy plasma arc is employed to melt a 16mm-thick aluminum alloy plate within a specialized assembly, in which the plate is butt-jointed to a transparent glass sheet, as illustrated in Fig.1. In this study, the transparent observation

window was optimized by employing quartz glass, which features a low thermal expansion coefficient and high mechanical strength. Most importantly, quartz glass has a high melting point of approximately 1750°C, which is significantly higher than that of GG17 glass (820°C) previously used in DC plasma arc welding [11]. This ensures excellent thermal stability under high-energy arc conditions and effectively prevents interference from glass melting, thereby enabling reliable in situ observation of keyhole and weld pool evolution. The high optical transparency of quartz glass further ensures a clear and undistorted view of the digging process, facilitating detailed analysis of keyhole and weld pool dynamics.

To capture the intricate dynamics of keyhole formation and weld pool behavior, a HAC-EX high-speed camera was utilized, operating at 2000 frames per second. In VPPAW of thick aluminum alloy, the high welding current generates an intensely bright arc, far exceeding the exposure range of the high-speed camera. To mitigate arc-induced overexposure and enhance image contrast, a 520 nm bandpass filter and a neutral density filter with 1 % light transmittance were placed in front of the camera lens. This optical configuration effectively suppresses arc glare while preserving fine details of the keyhole and weld pool. The high-speed camera, mounted orthogonally to the welding direction, captured the full thickness of the workpiece, enabling comprehensive observation of the entire penetration process. To further enhance the visibility of keyhole edges and weld pool boundaries, an auxiliary illumination system was implemented. A 520 nm wavelength laser with an output power of 40 W was positioned at the side of the workpiece, illuminating the observation area. This targeted lighting approach improves the detection of subtle shape variations, thereby increasing the accuracy of morphological analysis and ensuring reliable data acquisition for keyhole stability assessment.

The welding current was measured using a current transformer integrated with an Advantech USB-4716 high-speed data acquisition card, operating at a sampling frequency of 10,000 Hz. This configuration ensured high temporal resolution, allowing for precise synchronization between the electrical signal and high-speed visual data. The welding current waveform corresponding to optimized seam formation parameter is depicted in

Fig. 2a, based on which the digging process are classified into four distinct stages: the preheating stage (I_p), cleaning stage (I_c), current rising stage (I_r), and stable welding stage (I_s). In I_p , the arc ignition is initiated with a DC of 80 A (I_0) and plasma gas flow rate of 0.8 L/min (R_0), progressively increasing the workpiece temperature and establishing a thermal foundation necessary for keyhole formation. The subsequent I_c maintains the current at 80 % of I_s , while retaining the plasma gas flow rate at 0.8 L/min (R_c), enabling a controlled transition from DC to alternating current (AC). This stage enhances arc stability and improves the removal of oxide layers from the aluminum surface. In I_r , both the welding current and plasma gas flow rate are rapidly increased, further intensifying the thermal energy input. The final I_s is characterized by the peak welding current of 420 A and a plasma gas flow rate of 1.6 L/min (R_n), as illustrated in Fig. 2a, b. Additionally, the AC operates in two phases: Electrode Negative (EN) and Electrode Positive (EP), both maintaining equal current magnitudes with a frequency ratio of 20:5. This dynamic polarity shift enhances arc stability and improves oxide layer removal, which is particularly beneficial for welding aluminum alloys. An enlarged view of the welding current during the stable welding stage is provided in Fig. 2c, offering a detailed representation of the current fluctuations.



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Fig. 2. Welding current and plasma gas flow rate during the digging process. (a) Measured current waveform, (b) Plasma gas flow rate, (c) Enlarged view of the welding current during the stable welding stage.

In VPPAW of thick aluminum alloys, defects such as undercutting, incomplete penetration, and keyhole collapse are commonly observed, particularly concentrated during the digging stage. To systematically analyze the digging process and identify key influencing factors, a series of controlled experiments were conducted. As summarized in Table 1, five distinct parameter sets were designed to evaluate their respective impacts on the digging dynamics. Among them, the optimized seam formation parameter serves as the standard process, as shown in Fig. 2a. Given the complex interplay of various factors

of the welding parameter, adjustments are primarily focused on modulating the energy and force outputs of the arc throughout the entire digging stage. Parameter 2: increasing the plasma gas flow rate enhanced arc force, destabilizing the weld pool and resulting in undercut defects. Parameter 3: elevating the welding current intensified keyhole formation, leading to excessive penetration, keyhole collapse, and cutting. Parameter 4: reducing the welding current weakened keyhole formation, resulting in insufficient keyhole formation and irregular weld bead formation. Parameter 5: Lowering the plasma gas flow rate diminished arc constriction, causing incomplete penetration. These selected parameters comprehensively cover typical welding scenarios and defect types, providing a representative basis for analyzing the digging process.

Table 1. Typical welding parameters designed in VPPAW process.

Parameter no.	I_0 (A)	$R_0 = 0.5 * R_n$ (L/min)	$I_c = 0.8 * I_s$ (A)	$R_c = R_0$ (L/min)	I_s (A)	R_n (L/min)
1 (standard)	80	0.8	336	0.8	420	1.6
2	80	0.9	336	0.9	420	1.8
3	80	0.8	360	0.8	450	1.6
4	80	0.8	312	0.8	390	1.6
5	80	0.7	336	0.7	420	1.4

2.2. Keyhole and weld pool boundary tracking utilizing SegFormer

Extensive boundary data of the keyhole and weld pool were captured through high-speed imaging. To ensure efficient, accurate, and uniform extraction of these dynamic features, a deep learning-based image processing approach utilizing the SegFormer architecture is developed. The gas-liquid interface between the plasma arc and the molten metal is defined as the keyhole boundary, whereas the liquid-solid interface between the molten

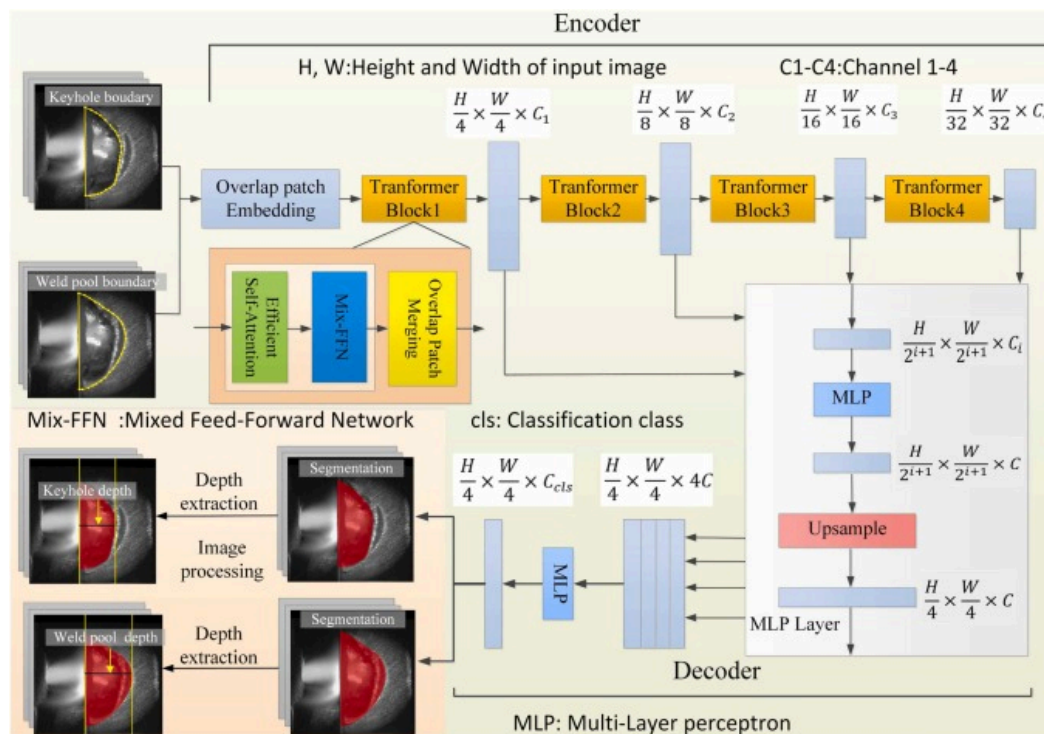
metal and the solid aluminum alloy is termed the weld pool boundary, as shown in [Fig. 3](#).

By utilizing a limited amount of labeled data, the evolutionary pattern of the digging process is effectively reconstructed. The detailed procedure for tracking the keyhole and weld pool depth is outlined as follows:

- (1) Construction of keyhole and weld pool boundary datasets. Given the absence of publicly available datasets specifically dedicated to the keyhole and weld pool boundary dynamics in VPPAW, a dedicated dataset was systematically constructed by acquiring the high-speed imaging data captured during the digging stage. The dataset incorporated different welding parameters to enhance the generalizability and robustness of subsequent analytical models. To further improve the accuracy of keyhole and weld pool boundary detection, the digging process was systematically segmented into discrete temporal intervals. A representative subset of images was selected from each interval to ensure comprehensive and unbiased data coverage throughout the entire sequence of the digging process.
- (2) Image Annotation. The Labelme tool offers flexibility in polygon-based segmentation, high annotation precision, and consistency in boundary delineation, making it well-suited for complex image labeling tasks [\[42\]](#). Therefore, the keyhole and weld pool boundaries in raw high-speed images were systematically annotated, as illustrated in [Fig. 3](#). Although the Labelme is a manual annotation tool, a highly detailed and meticulous boundary annotation process was employed. The annotations were carefully reviewed multiple times to minimize errors and ensure precision. Additionally, the data selected for labeling was deliberately chosen to exhibit significant variation, which helps to avoid overfitting when used for training the Segformer model. Through precise annotation of these boundaries, a dataset comprising 500 meticulously labeled images was established, serving as a

robust foundation for subsequent quantitative analysis and the development of machine learning-based predictive models.

- (3) Model training and prediction. Leveraging the advanced image segmentation capabilities of SegFormer, the annotated dataset is subsequently fed into the model for training and predicting. As illustrated in Fig. 4, the SegFormer architecture adopts an encoder-decoder structure, specifically designed to achieve high-precision segmentation of keyhole and weld pool boundaries [43]. The training and prediction processes are executed on an NVIDIA RTX 3090 GPU, ensuring computational efficiency and accelerated model convergence. By employing the SegFormer model, the keyhole and weld pool regions throughout the digging process are rapidly and accurately segmented.

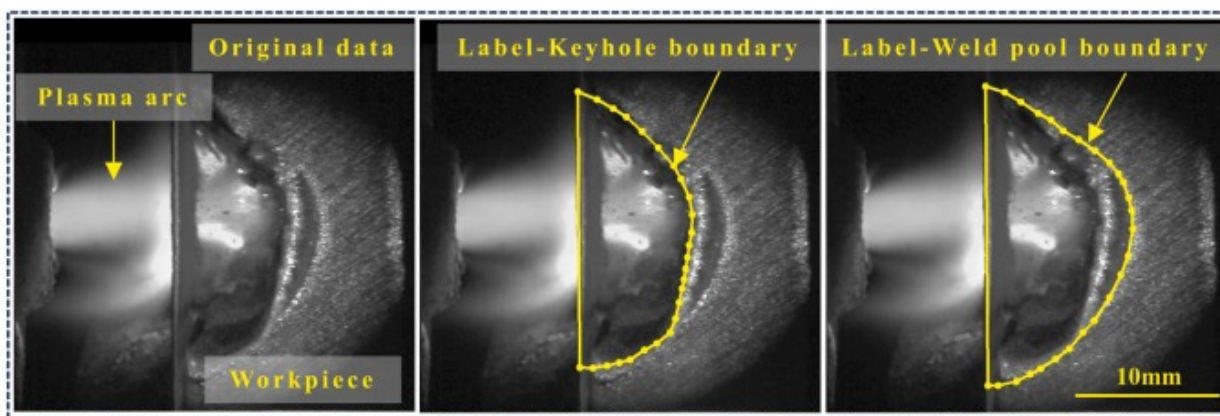


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Fig. 4. Keyhole and weld pool depths extraction using optimized SegFormer architecture.

- (4) Depth extraction. The depths of the keyhole and weld pool reflect the combined effects of force distribution, thermal energy input, and molten metal fluid dynamics, and thus serve as key geometric descriptors of the welding process. To elucidate the morphological evolution of the keyhole and weld pool over time, this study utilizes quantitative measurements of both keyhole and weld pool depths. To enhance the accuracy of the segmentation results, the keyhole and weld pool regions are firstly refined using morphological operations, effectively mitigating image noise and preserving structural integrity. Subsequently, edge detection techniques are applied to precisely delineate the keyhole and weld pool contours [44]. The depth of these boundaries is quantitatively determined using the horizontal maximum value method, ensuring a robust and consistent representation of depth variations throughout the digging process.



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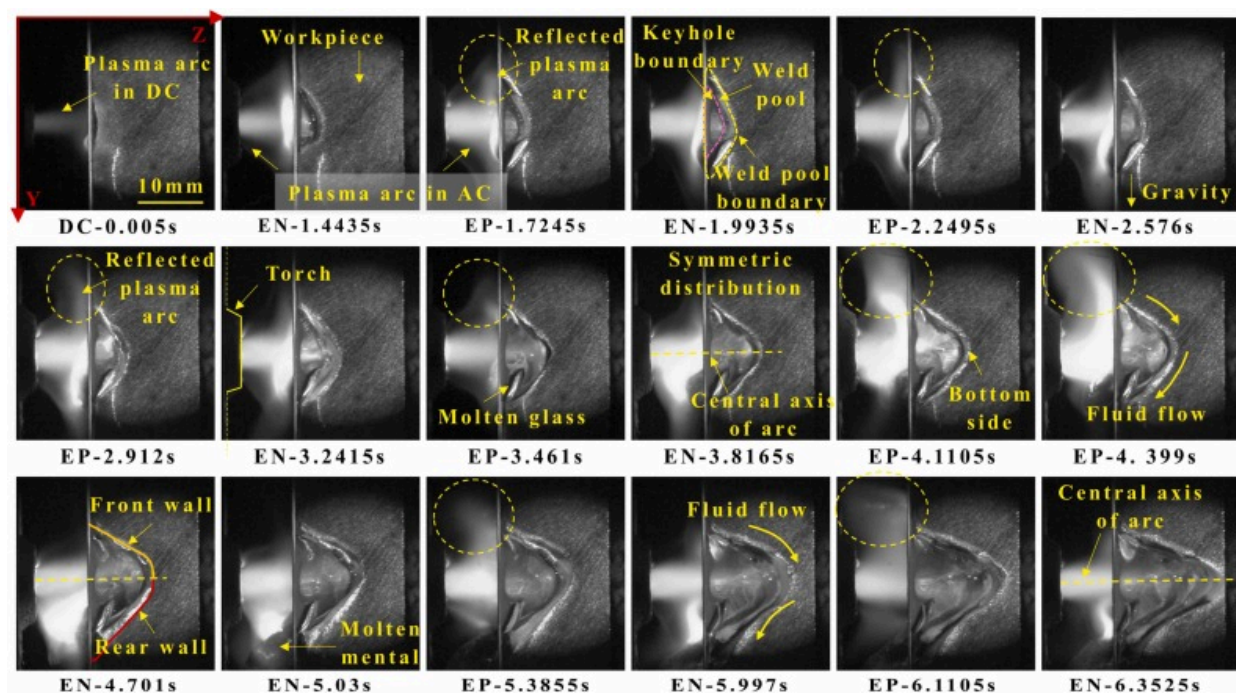
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Fig. 3. Annotation of keyhole and weld pool boundaries using the Labelme tool.

3. Results and discussion

3.1. Dynamic evolution of internal keyhole and weld pool under standard welding parameter

Fig. 5 illustrates the keyhole and weld pool formation process under standard welding parameter (parameter 1). The transparent observation setup facilitates a clear visualization of keyhole and weld pool dynamics. The quartz glass exhibits minimal deformation, ensuring that the captured images accurately reflect the keyhole and weld pool evolution. The initial reference time is defined as the onset of the preheating stage, as shown in Fig. 2a. The lower section of each subfigure indicates the corresponding welding phase (EN or EP) and the specific time at which the data was recorded. The Y-direction is defined as perpendicular to the Z-direction and oriented opposite to the welding direction. The Z-direction represents the depth axis along which the keyhole penetrates and evolves as the melting process advances. The front keyhole wall corresponds to the upper boundary of the keyhole, located ahead of the welding torch, whereas the rear keyhole wall is positioned behind it, as illustrated at 4.701 s.



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Fig. 5. Keyhole and weld pool formation process under standard welding parameter.

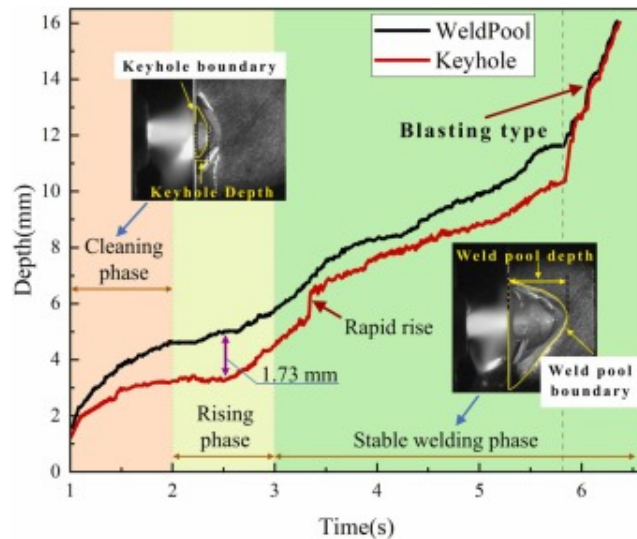
During the preheating stage, when operating with DC, the plasma arc exhibits low brightness due to the limited current magnitude, as shown in Fig. 5 at 0.005 s. As the current increases and transitions into the AC phase, the arc brightness intensifies significantly. Additionally, the arc demonstrates distinct characteristics during the EN and EP phases, with the cross-sectional area of the arc in the EP phase being substantially larger than that in the EN phase. According to Xu et al., arc energy is more concentrated during the EN phase, generating a higher arc force, whereas in the EP phase, the arc expands, enhancing the effectiveness of cathodic cleaning [45]. Furthermore, an intriguing phenomenon emerges: a distinct arc reflection is observed in the front wall of

the keyhole during the EP phase, while this reflection is absent in the EN phase. This reflection serves as a key indicator for differentiating between the EN and EP phases.

The plasma-induced pressure displaces the liquid metal, while gravitational force directs its downward movement within the weld pool, collectively contributing to keyhole formation. The weld pool is the region of molten metal situated between the weld pool boundary and the keyhole boundary. As the welding process progresses, the high-energy arc continuously melts the aluminum alloy, leading to a dynamic evolution in the morphologies of the weld pool and keyhole. The keyhole boundary is delineated by the purple curve, while the weld pool boundary is represented by the yellow curve, as illustrated in Fig. 5 at 1.9935 s. Prior to 3.8 s, the keyhole and weld pool exhibit an approximately symmetrical distribution along the Y-direction, with the central arc axis serving as the reference line. Subsequently, the lower portion of the keyhole and weld pool undergoes a gradual expansion compare to the upper portion. Upon full penetration, the disparity in area between the upper and lower regions reaches its maximum.

Fig. 6 quantitatively illustrates the evolution of keyhole and weld pool depths under parameter 1. During the DC phase, the plasma arc exhibits relatively low intensity, resulting in negligible changes in both keyhole and weld pool depths. As a result, this study primarily focuses on the depths evolution during the cleaning, rising, and stable welding phases. During the cleaning phase, where the current has reached 80 % of the stable welding current, the high-energy arc rapidly melts the metal. The corresponding weld pool depth displays a logarithmic growth trend, characterized by a steep initial increase followed by a gradual deceleration. Upon transition into the rising phase, despite continuous increases in both welding current and plasma gas flow rate, the rate of weld pool deepening diminishes. During the stable welding phase, where both welding current and plasma gas flow rate attain their peak values, the depth evolution becomes increasingly complex. Initially, the depth follows a parabolic growth trajectory, which is subsequently interrupted by a quasi-steady stage during which the depth remains relatively stable. A second parabolic growth segment then emerges, though at a reduced rate, and is followed by another brief quasi-steady state. Notably, as thermal energy

continues to accumulate within the weld pool, a rapid increase of 4.37 mm in depth occurs within a span of 0.527 s, signifying a blasting-type penetration (BP) event. In summary, the evolution of the weld pool depth undergoes oscillatory digging, consisting of three alternating cycles of rapid growth and quasi-steady behavior, ultimately culminating in a BP event.



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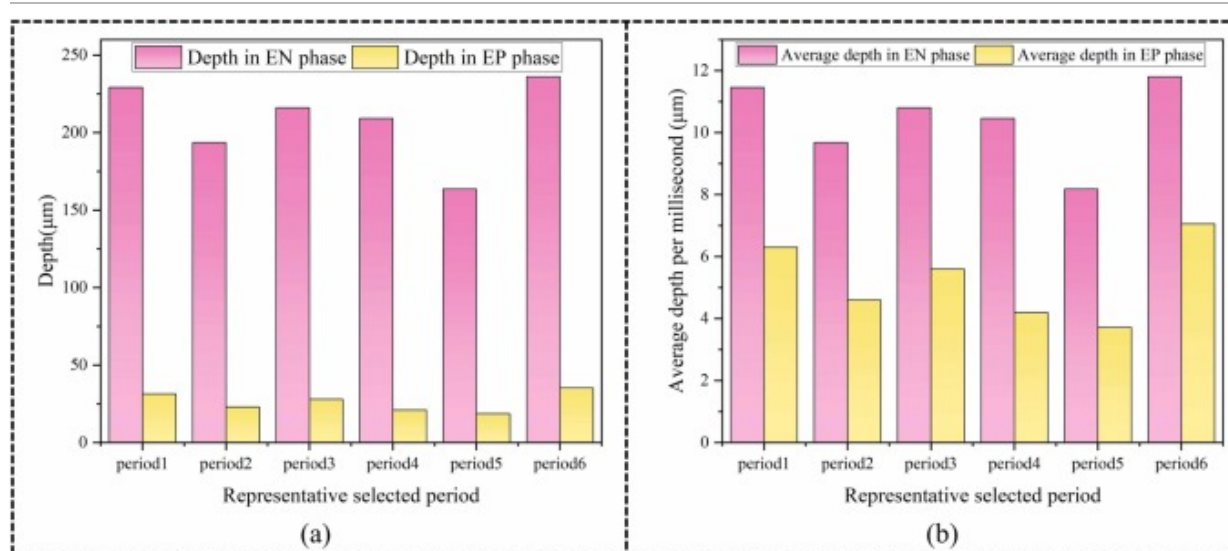
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Fig. 6. Evolution of keyhole and weld pool depths under parameter 1.

The evolution of keyhole depth closely mirrors that of the weld pool depth. The difference value between the keyhole depth and the weld pool depth defines the weld pool thickness at the bottom of the keyhole (WPTBK), serving as a critical indicator of energy transfer and fluid dynamics within the weld pool. During the cleaning phase, the keyhole depth reaches a quasi-steady state approximately 0.5 s earlier than the weld pool depth. In the subsequent rising phase, the WPTBK reaches a peak value of 1.73 mm at 2.58 s. Notably, during the subsequent stable welding phase, the keyhole depth undergoes a sudden and rapid increase, which is a unique feature in VPPAW of thick aluminum alloy.

Thereafter, the keyhole depth transition to a gradual deepening trend. At the moment of blasting penetration BP, the depths of the keyhole and weld pool nearly converge, with the weld pool boundary reaching the full plate thickness at 6.357 s.

The BP process occurs within a brief time interval and is characterized by a sharp increase in both keyhole and weld pool depths, significantly influencing weld quality. For detailed analysis, the weld pool depth growth per polarity cycle is examined over six discrete phases, encompassing both EN and EP phases. As shown in Fig. 7a, these intervals are defined as: period 1 (5.900–5.925 s), period 2 (6.0–6.025 s), period 3 (6.05–6.075 s), period 4 (6.15–6.175 s), period 5 (6.2–6.225 s), and period 6 (6.3–6.325 s). The measured weld pool depth growths during the EN phase are 229.2 μm , 193.6 μm , 216.2 μm , 209.2 μm , 163.6 μm , and 236.2 μm , respectively. In contrast, the corresponding growths during the EP phase are 31.5 μm , 23.0 μm , 28.0 μm , 21.0 μm , 18.6 μm , and 35.3 μm . Given the EN and EP durations of 20 ms and 5 ms per polarity cycle, the average depth growths per millisecond for the EN phase are calculated as 11.46 μm , 9.68 μm , 10.81 μm , 10.46 μm , 8.18 μm , and 11.81 μm . For the EP phase, the corresponding values are 6.30 μm , 4.60 μm , 5.60 μm , 4.20 μm , 3.72 μm , and 7.06 μm , as illustrated in Fig. 7b. The resulting EN-to-EP depth ratios are 1.82, 2.10, 1.93, 2.49, 2.20, and 1.67, respectively. The results reveal that the oscillatory behavior of the weld pool depth occurs on the millisecond-level polarity cycles. As previously demonstrated, the oscillatory behavior also persists throughout the entire digging process. These findings highlight a critical phenomenon: during the BP stage, the weld pool depth undergoes oscillatory fluctuations at both macroscopic (entire digging duration) and microscopic (individual polarity cycle) temporal scales. The dual-scale oscillation reflects the intricate interplay between cyclic energy input and the fluid flow dynamics of the molten metal. In particular, the higher penetration efficiency observed during the EN phase is attributed to its stronger arc pressure and higher thermal energy density compared to the EP phase [45]. The cyclic thermal and mechanical fluctuations underscore the inherently unsteady and dynamic nature of the digging process in VPPAW of thick aluminum alloy.



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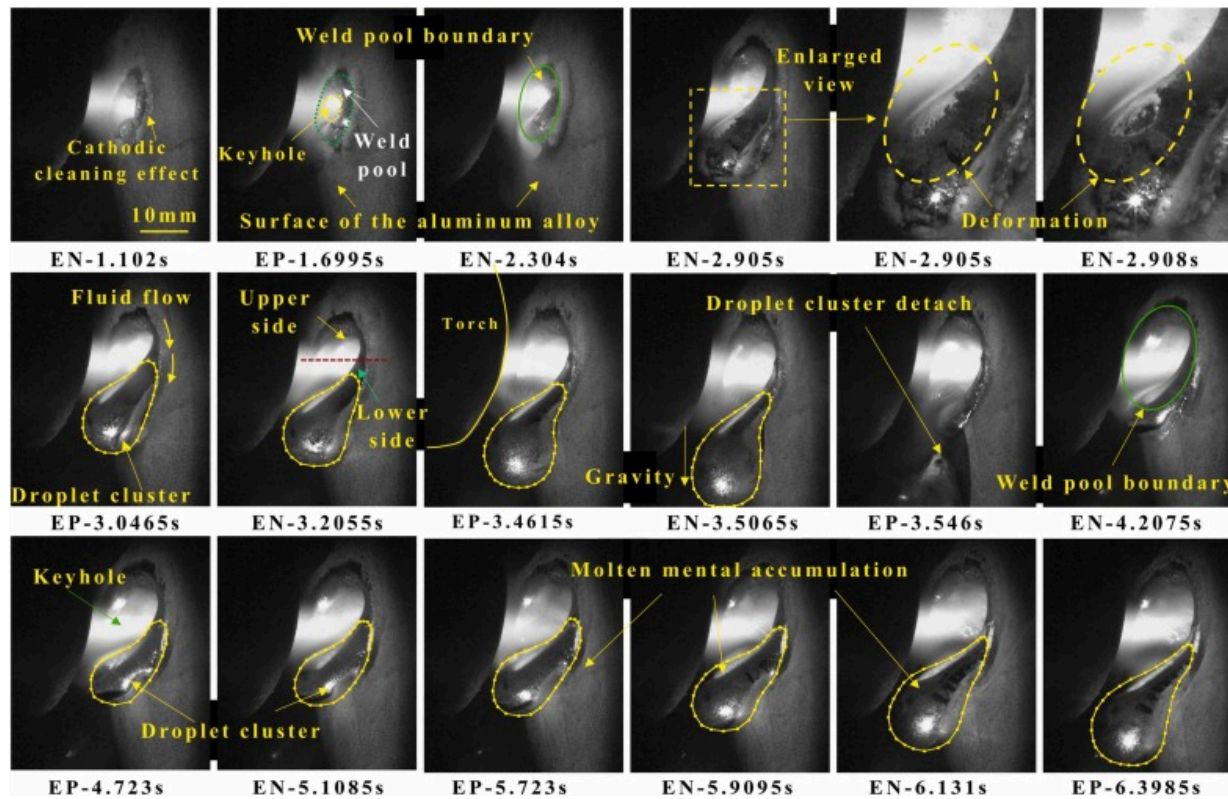
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Fig. 7. Weld pool depth growth during the representative welding period (period1: 5.9–5.925 s, period2: 6.0–6.025 s, period3: 6.05–6.075 s, period4: 6.15–6.175 s, period5: 6.2–6.225 s, period6: 6.3–6.325 s). (a) Depth growth during the corresponding Electrode Negative (EN)/Electrode Positive (EP) phase (Each period is subdivided into an EN phase and an EP phase), (b) Average depth growth per millisecond during each EN/EP phase (an EN phase lasts 20 ms, while an EP phase lasts 5 ms).

3.2. Dynamic evolution of surface keyhole and weld pool under standard welding parameter

To investigate the dynamic evolution of the keyhole and weld pool during actual welding, the experimental setup was optimized by replacing the glass-aluminum butt joint with a full aluminum alloy plate. Furthermore, the high-speed camera was repositioned to capture a lower-angle side view, enabling continuous observation of the keyhole morphology throughout the penetration process. The specific shooting angle and position of the high-speed camera are illustrated in Fig. 10a. Fig. 8 presents the evolution of the

surface keyhole and weld pool under welding parameter 1, revealing distinct digging stage of formation and development. At 1.102 s, during the initial cleaning stage, the cathodic spot cleaning region is distinctly observed. In addition, the application of a high-energy arc induces localized melting of the metal, thereby facilitating the formation of the keyhole and weld pool. In the early penetration stage (prior to 2.304 s), the limited volume of molten metal minimizes gravitational influence, resulting in molten material accumulation primarily around the keyhole. However, as the arc energy input continues and the weld pool volume increases, gravitational forces become increasingly significant, gradually driving the downward flow of molten metal and promoting its accumulation at the lower side of the keyhole. Fig. 8 presents enlarged views of the weld pool at 2.905 s and 2.908 s, revealing a distinct morphological deformation within a mere 0.003 s interval. High-speed imaging captured this rapid transformation, highlighting the pronounced influence of arc-induced forces on the molten metal. These transient deformations exemplify the intense dynamic disturbances imposed by the arc, which significantly affect the evolution of both keyhole and weld pool geometries. Notably, the dual-scale oscillation observed in weld pool depth, which appears both at the macroscopic level and within individual polarity cycle, is attributed to the intrinsic oscillatory behavior of the arc force.



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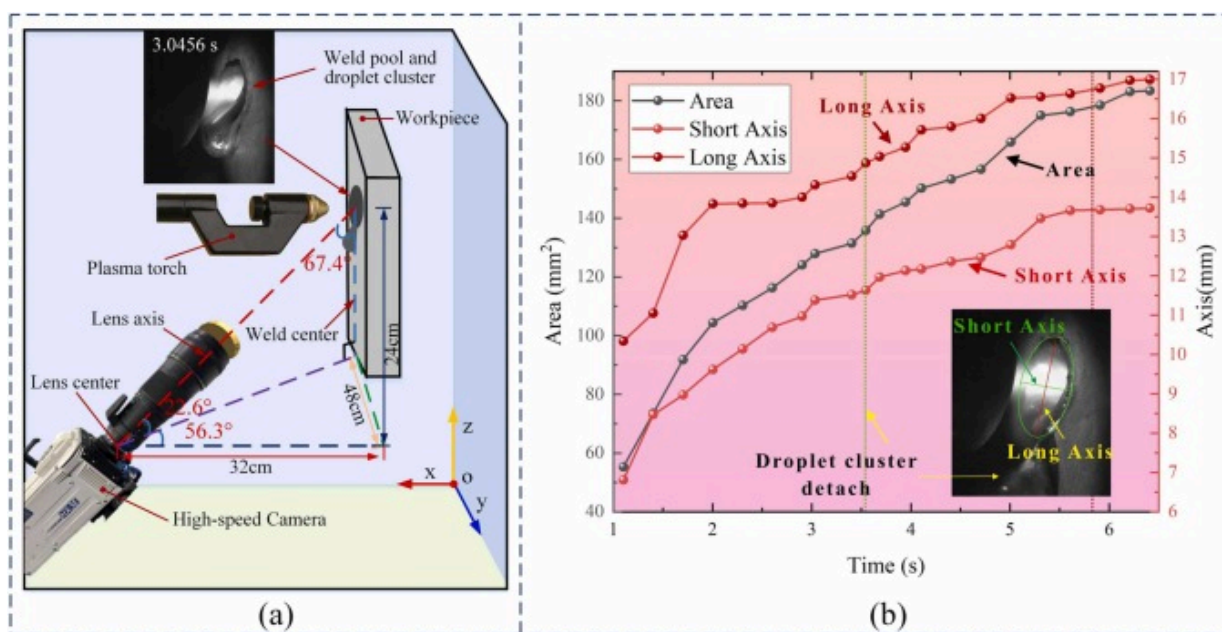
Fig. 8. Surface keyhole and weld pool formation process under standard welding parameter.

As the digging process advances, an increasing volume of molten metal accumulates at the lower side of the keyhole, gradually forming a substantial molten metal droplet cluster. Meanwhile, the upper side of the keyhole shows a marked absence of molten metal accumulation, as evidenced in Fig. 8 at 4.4075 s. Driven by the combined effects of gravity and arc-induced oscillatory forces, the droplet cluster eventually detaches and falls, as illustrated in Fig. 8 at 3.546 s. The detachment of droplet cluster during penetration represents a distinctive characteristic in VPPAW of thick aluminum alloy,

arising from the generation of a large-volume weld pool. Following detachment, a new droplet cluster gradually forms as molten metal continues to accumulate. Ultimately, full penetration is achieved as the arc successfully traverses the thickness of the aluminum alloy. Accordingly, the dynamic surface behavior of the keyhole and weld pool are classified into four characteristic modes: (1) molten metal surrounding the keyhole, (2) molten metal flowing downward and accumulating to form a droplet cluster, (3) detachment of the accumulated droplet cluster, and (4) reformation and growth of a new droplet cluster.

As illustrated in [Fig. 8](#) at 2.304 s and 3.546 s, the surface weld pool boundary presents a distinct elliptical morphology, as captured from the specific camera observation angle. To quantitatively analyze its geometric characteristics, an ellipse fitting algorithm was applied, followed by calculations of the area, long axis, and short axis dimensions. [Fig. 9b](#) shows the dynamics of surface weld pool boundary dimensions during the VPPAW digging process. The initial rapid expansion of the surface weld pool boundary is driven by the intense thermal and mechanical effects generated of the arc at a high welding current of 420 A, which induce intense melting and fluid flow. Within 2 s, the long axis of the weld pool rapidly extends to 14 mm. Similarly, the short axis and area increase to 12 mm and 140 mm², respectively, by approximately 3.7 s. However, as the penetration process progresses, the surface weld pool enters a phase of gradual expansion. This behavior is attributed to the variable polarity plasma arc functioning as a constricted arc [\[46\]](#), which limits the volume of the plasma arc. Once the surface weld pool attains a critical dimension, the arc-induced confinement significantly suppresses further growth of the weld pool boundary. In contrast, a substantial portion of the energy is redirected towards deepening the keyhole and facilitating molten metal flow along the thickness direction. In the final stage, beyond 5.5 s, the majority of arc energy is primarily consumed in melting the bottom and sidewalls of the keyhole, further limiting surface weld pool boundary dimensions expansion. These observations align with the findings of Zhou et al. [\[11\]](#), who reported analogous dynamic behaviors in weld pool width during

the digging process. Upon full penetration, the long axis, short axis, and area reach 16.9 mm, 13.7 mm, and 183.4 mm², respectively.



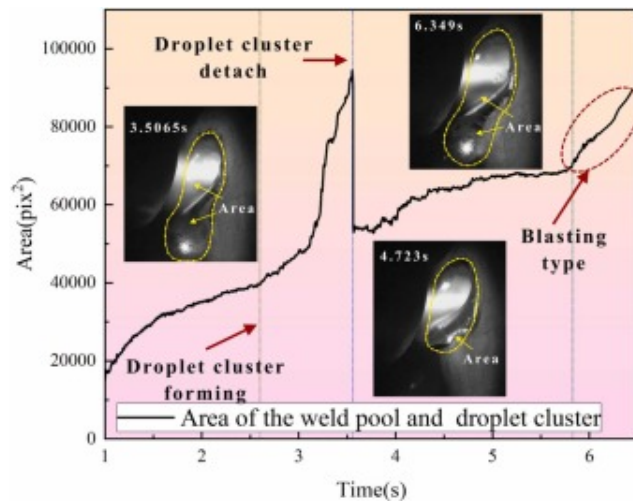
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Fig. 9. Shooting position and angle of the monitoring system for the surface keyhole and weld pool formation process, and dynamic evolution of surface weld pool boundary dimensions during the digging process. (a) Shooting position and angle of the monitoring system for the surface keyhole and weld pool formation process. (b) Dynamic evolution of surface weld pool boundary dimensions during the digging process.

The internal dynamics of the keyhole and weld pool play a crucial role in the formation of high-quality weld seams. However, monitoring these internal characteristics in real time during welding is challenging. Therefore, establishing a correlation between surface features and internal behaviors is essential for optimizing the welding process. As illustrated in Fig. 9, surface weld pool boundary dimensions alone are inadequate for accurately reflecting the internal evolution of the keyhole and weld pool. Although the

droplet cluster provides valuable insights into internal dynamics of the weld pool, its continuously varying shape poses significant challenges for rapid and precise extraction, as demonstrated in Fig. 8. To address this limitation, this study proposes a fusion-based extraction method that integrates information from both the weld pool boundary and the droplet cluster boundary. Fig. 10 depicts the dynamic evolution of the surface weld pool and droplet cluster (SWPDC) area throughout the digging stage. Consistent with the keyhole depth extraction methodology, the SWPDC boundary is initially annotated using the Labelme tool. Subsequently, the SegFormer architecture is employed to achieve automated segmentation and extraction of the SWPDC region across the entire digging sequence. Following segmentation, the SWPDC area is quantitatively computed using image processing techniques. Given the irregular shape of the SWPDC boundary, size conversion easily introduces errors. Therefore, this study employs pixel values to directly quantify evolution of SWPDC area. The original image resolution of 654×790 pixels ensures sufficient detail for precise boundary extraction and analysis.



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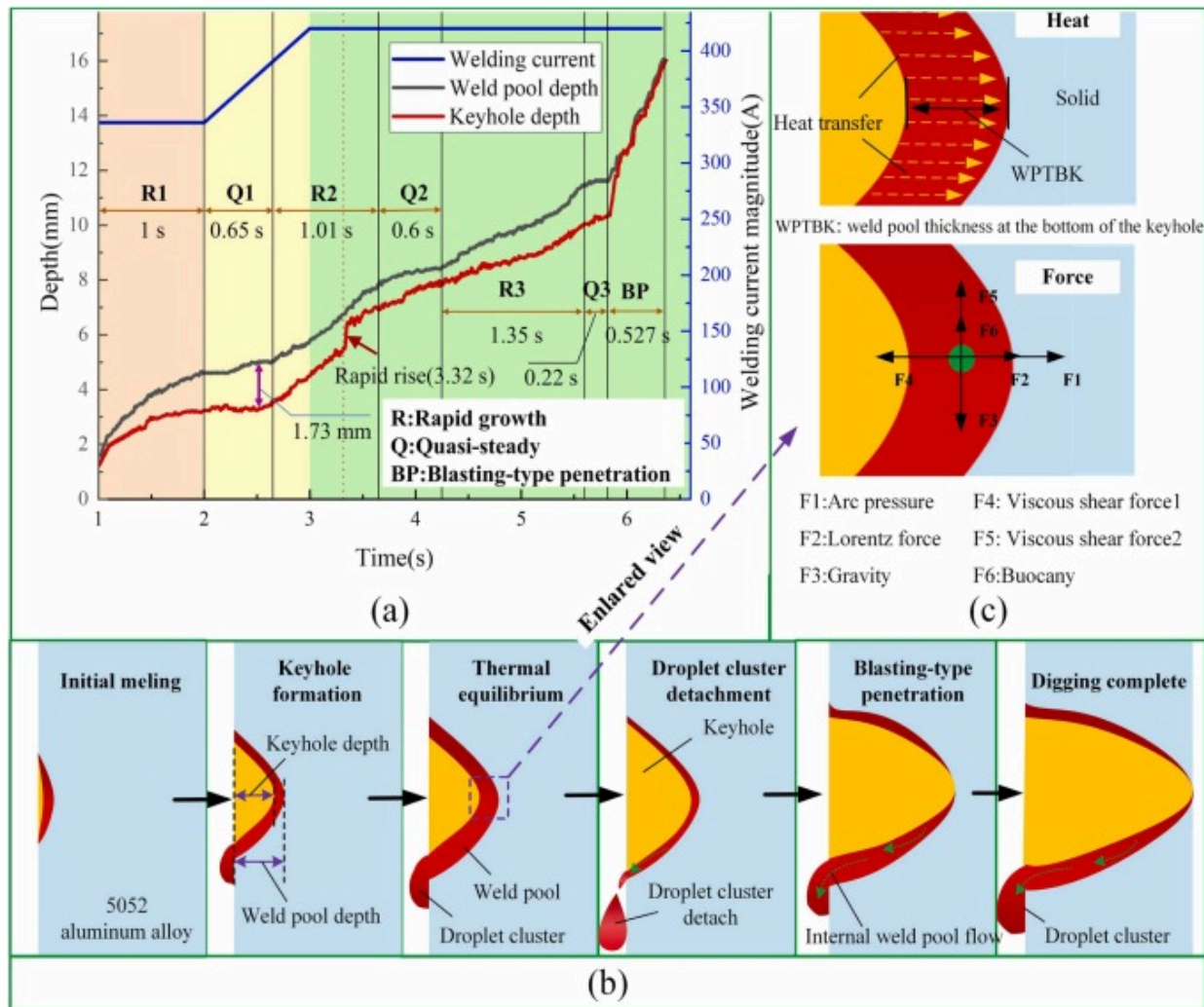
Fig. 10. Dynamic evolution of the surface weld pool and droplet cluster area during the digging process.

Fig. 10 illustrates the dynamic evolution of the SWPDC area throughout the digging process. Initially, the droplet cluster is absent, with only the weld pool boundary visible. The weld pool boundary gradually expands under the combined action of thermal and mechanical effects exerted by the high-energy arc. As the keyhole deepens, molten metal accumulates at the lower side of the workpiece, leading to the formation of a droplet cluster around 2.6 s. Furthermore, molten metal continuously flows out from the interior of the keyhole, driving the rapid expansion of the SWPDC area, as shown in **Fig. 10**. Simultaneously, the keyhole depth increases progressively, as depicted in **Fig. 6**. The phenomenon is facilitated by the emergence of the droplet cluster, which enhances the transport of internal molten metal to the surface of the workpiece, thereby improving arc melting efficiency in the thickness direction of the workpiece. At 3.546 s, the droplet cluster detaches abruptly, resulting in a sudden decline in the SWPDC area. Almost simultaneously, at 3.35 s, the keyhole depth experiences a rapid increase, as shown in **Fig. 6**. Subsequently, the accumulation of molten metal on the surface leads to another phase of SWPDC area growth. Notably, after 5.83 s, the SWPDC area exhibits a sharp expansion, corresponding to the BP phenomenon occurring within the keyhole, as shown in **Fig. 6**, **Fig. 10**. This rapid expansion results from the intense metal melting during the BP phase, which drives the swift flow of molten metal from the keyhole interior to the surface droplet cluster. The rapid expansion of the SWPDC area serves as a reliable external indicator of internal weld pool BP dynamics. These observations confirm that the internal evolution of the keyhole and weld pool morphology can be effectively characterized through the external behavior of the SWPDC area, offering valuable insight for real-time monitoring and control during the digging process.

3.3. Mechanism of keyhole and weld pool dynamic behaviors

Unlike the linear increasement in keyhole and weld pool depth typically observed in DC plasma arc welding [11], the oscillatory and BP behaviors during the digging stage of VPPAW are unique phenomena. **Fig. 11a** illustrates the detailed segmentation of the digging process into representative stages, namely rapid growth (R1-R3), quasi-steady (Q1-Q3), and BP. Correspondingly, **Fig. 11b** shows the typical evolution characteristics. In

the initial rapid growth stage (R1), despite the current reaching 80 % of the stable welding value, the bottom of the keyhole is positioned close to the torch, resulting in a concentrated thermal effect and a steep increase in weld pool depth. Simultaneously, the arc pressure is considerable, exerting a significant force on the molten metal between the keyhole and the weld pool boundary. The arc pressure inhibits further downward extension of the keyhole, causing an increase in the WPTBK. The arc heat then propagates through the weld pool, melting the solid metal via conduction. As the WPTBK thickens, heat dissipation increases, resulting in a reduced rate of weld pool deepening. Additionally, the 5052 aluminum alloy used in this study possesses a high thermal conductivity of $138 \text{ W/(m}\cdot\text{K)}$ [47], which is substantially greater than that of stainless steel ($13 \text{ W/(m}\cdot\text{K)}$ [48]) used in comparable DC plasma arc welding [11]. The combined effects of the thickened WPTBK and the superior thermal conductivity of aluminum drive the system into the first quasi-steady stage (Q1), where a dynamic equilibrium between arc heat input and thermal dissipation results in a relatively stable weld pool depth.



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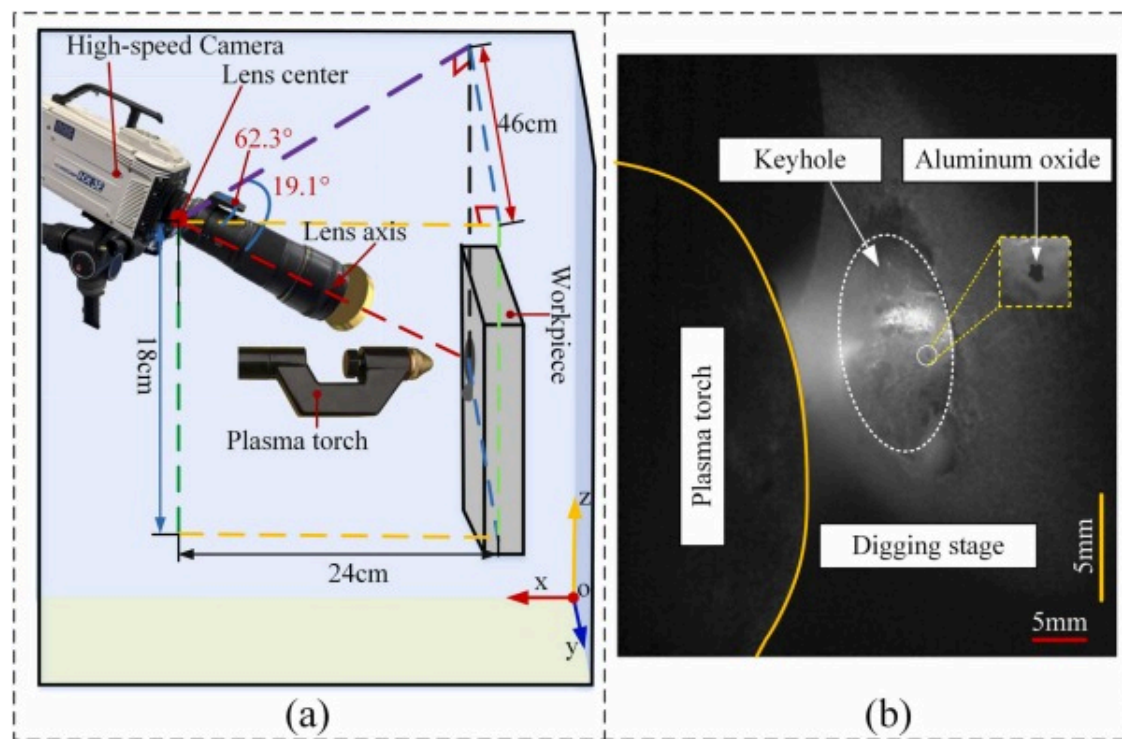
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Fig. 11. Formation mechanism of keyhole and weld pool during digging process. (a-b) Temporal evolution of keyhole and weld pool depths with segmentation into representative stages. (c) Schematic illustrations of heat transfer and driven forces within the weld pool.

As illustrated in Fig. 11c, the internal weld pool is subjected to multiple driving forces, primarily including arc pressure (F_1), Lorentz force (F_2), gravity (F_3), viscous shear force (F_4 , F_5) and buoyancy (F_6) [49]. Once the WPTBK accumulates to 1.73 mm, the gravitational force (F_3) becomes sufficiently large, exceeding the combined effects of the viscous shear forces (F_5) and buoyancy (F_6) ($F_3 > F_5 + F_6$). This disruption in force equilibrium causes the molten metal to flow towards the rear wall of the keyhole, promoting the transition from Q1 to R2. Additionally, to ensure welding stability for medium-thickness aluminum plates, it is common practice to apply a higher current in the EP phase, usually by approximately 30 A, compared to the EN phase [14], [24]. However, in this study, the current magnitudes for the EP and EN phases were kept identical, inevitably introducing periodic fluctuations in arc force. As demonstrated in Fig. 7, the oscillatory behavior of the weld pool depth occurs on the millisecond-level polarity cycles, originating from fluctuations in arc energy. These rapid, cyclic fluctuations in energy input generate localized thermal gradients within the weld pool, which subsequently result in transient variations in the viscosity and flow velocity of the molten metal. Moreover, these energy fluctuations cause the molten metal to undergo dynamic redistribution, which induces localized turbulence at the bottom of the keyhole. Over time, the cumulative effects of these oscillatory fluctuations progressively exacerbate the thermal disturbance established during the Q1 stage. In conjunction with gravitational force, these fluctuations promote the outward flow of molten metal from the bottom of the keyhole, resulting in progressive thinning of the WPTBK. Consequently, the reduced thermal resistance enables the arc energy to more effectively concentrate at the keyhole bottom, enhancing local melting and driving the transition into the second rapid growth stage (R2).

The surface flow behavior of the weld pool provides critical insights into the thermo-fluid coupling and arc-pool interaction mechanisms during the VPPAW process. In thick plate welding, the high welding current and the associated intensive cooling requirements result in a relatively large torch head, which consequently restricts the available observation angle. Therefore, a dedicated experiment was designed to capture the surface

flow dynamics of the weld pool along the side and rear keyhole walls, as illustrated in Fig. 12. Fig. 12a presents the schematic of the surface flow observation system, detailing the shooting position and angle of the high-speed camera. The camera setup and laser light source were identical to those used in the transparent keyhole and weld pool monitoring system. Fig. 12b presents a schematic representation of aluminum oxide serving as tracer particles within the weld pool, a method that has been widely employed to characterize convection patterns and velocity distributions in VPPAW [49]. It is worth noting that the length scale bars in the horizontal and vertical directions are provided separately according to the shooting angle of the high-speed camera.



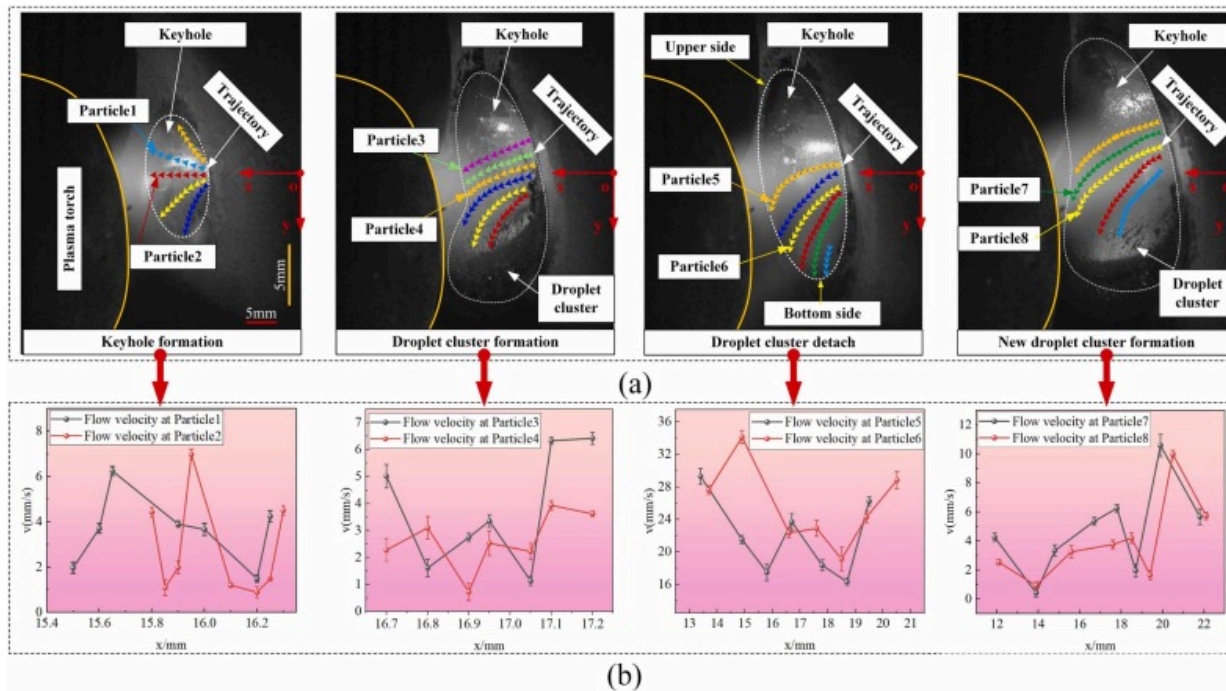
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Fig. 12. Surface weld pool flow monitoring system. (a) Schematic of shooting position and angle of the high-speed camera. (b) Aluminum oxide acting as tracer particle in the weld

pool.

Fig. 13 illustrates the surface flow behavior of the weld pool across multiple stages, including keyhole formation, droplet cluster formation, droplet cluster detachment, and subsequent new droplet cluster formation. Different colored arrows represent distinct tracer particles, as shown in Fig. 13a. During the initial digging stage, when the keyhole remains relatively small, the molten metal exhibits multidirectional convection driven by both arc heat and arc pressure. As the keyhole enlarges and deepens, droplet cluster gradually emerge. The tracer-particle trajectories show a consistent surface flow from the keyhole sidewall toward its bottom, which persists during droplet-cluster detachment and re-formation. Additionally, the flow velocities corresponding to the transition from keyhole formation to new droplet cluster formation were quantitatively analyzed, as presented in Fig. 13b. The coordinate system was established with its origin at the midpoint of the right boundary of the image, as indicated in Fig. 13a, with the x-axis directed to the left and the y-axis toward the bottom. The flow velocity was calculated as the vector magnitude from the x- and y-directional components, with two representative tracer particles selected at each stage for analysis. The results show that during the keyhole and droplet-cluster formation stages, the flow velocity ranges from 0 to 8 mm/s. As the new droplet cluster forms, the velocity increases slightly to 0–11 mm/s. In contrast, during the droplet-cluster detachment stage, the velocity rises sharply to 16–36 mm/s, significantly exceeding the velocities observed in the other three stages.



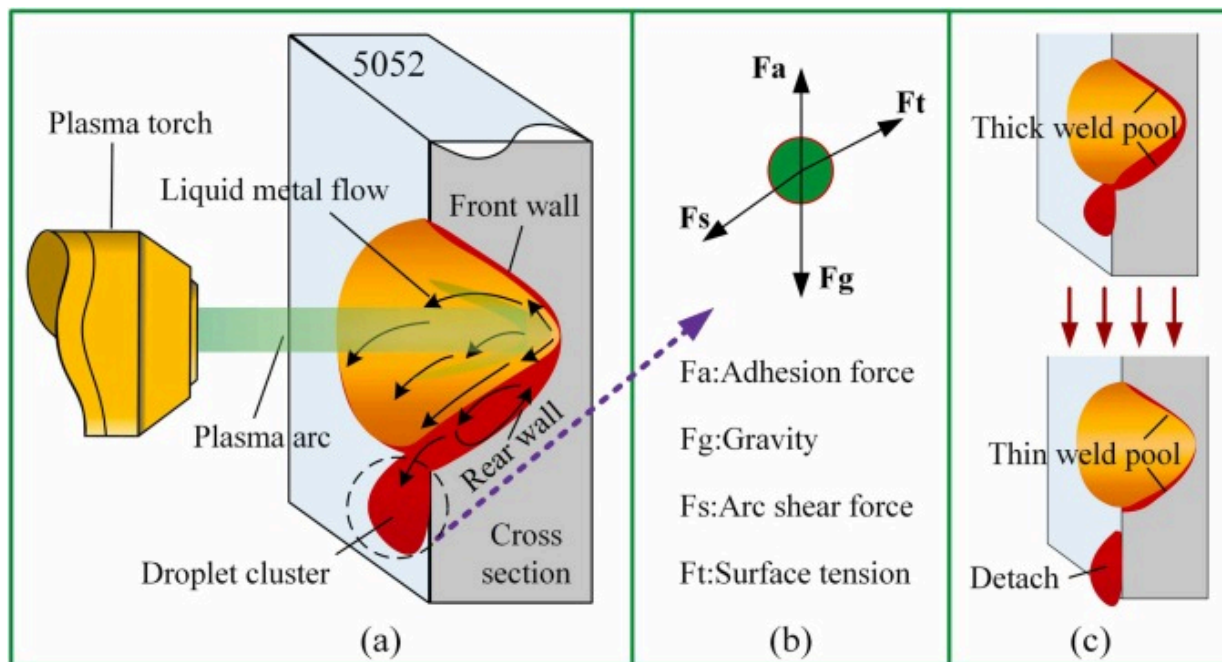
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Fig. 13. The surface flow behavior of the weld pool from the keyhole formation stage to the new droplet cluster formation stage. (a) The surface flow trajectory of the weld pool at different stages. (b) The corresponding flow velocity distributions at each stage.

Fig. 14a provides a schematic illustration of the fluid flow within the weld pool during the digging process. As the arc acts on the bottom of the blind keyhole, a strong reverse shear force is generated, driving the molten metal at the rear wall of the keyhole to rapidly flow toward the entrance side of the keyhole. Under the influence of gravity, molten metal accumulates at the rear wall of the keyhole, while a reflux phenomenon occurs within the weld pool [50]. The molten metal flows out from the interior of the keyhole, resulting in the formation of droplet cluster, as demonstrated in Fig. 8. The forces acting on the droplet cluster are depicted in Fig. 14b, primarily influenced by adhesion force (F_a),

gravity (F_g), arc shear force (F_s), and surface tension (F_t). The components of F_s and F_t in the vertical direction are represented by F'_s and F'_t , respectively. As the droplet cluster accumulates, it begins to move downward at an accelerated rate when the following condition is satisfied: $F_g + F'_s > F_a + F'_t$. By 3.38 s, the droplet cluster detaches from the weld pool boundary, triggering a rapid increase in keyhole depth. Notably, the detachment of the droplet cluster occurs approximately 0.2 s after the rapid increase in keyhole depth. This delay is attributed to the continuous outflow of molten metal from within the keyhole, combined with the accelerated motion of the droplet cluster prior to its detachment. Additionally, since the droplet cluster is physically bonded with the molten metal inside the keyhole, its detachment initiates the expulsion of molten material from the keyhole, driven by the concurrent effects of surface tension and gravity, as illustrated in Fig. 14c.



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Fig. 14. Schematic illustration of weld pool flow, driven forces on the droplet cluster, and the transient evolution of the droplet cluster with corresponding weld pool variations. (a) Fluid flow in the weld pool. (b) Driven forces on the droplet cluster. (c) Droplet cluster detachment and associated weld pool variations.

During the blind digging stage of VPPAW, the accumulation of thick molten metal at the bottom of the keyhole significantly impedes the digging process. The expulsion of molten metal from the keyhole is essential for efficient penetration, as it reduces resistance and enables deeper arc advancement. This process is enhanced by droplet cluster detachment, which assists in maintaining stable keyhole dynamics. While the outflow of molten metal from the keyhole interior is beneficial for the keyhole digging, excessive expulsion of molten material results in a reduced volume of molten metal within the keyhole, which can adversely affect the welding stability. As previously described, the VPPAW process are categorized into three distinct stages: the digging stage, the initial unstable stage, and the stable welding stage. During the transition from the digging stage to the initial unstable stage, the welding torch begins to traverse along the surface of the workpiece. As the torch advances, the plasma arc continuously melts the solid metal at the keyhole front wall, and the molten metal subsequently flows along the side walls toward the rear wall to facilitate keyhole closure [49]. However, in the digging stage, a significant amount of molten metal is expelled from the keyhole, leaving insufficient material to adequately replenish the rear wall. This imbalance hinders complete keyhole closure and weld seam formation, often resulting in defects such as undercuts and lack of fusion. The inability to maintain an optimal molten metal balance during the digging stage is a primary factor contributing to the typically narrow process parameter in VPPAW of thick workpiece.

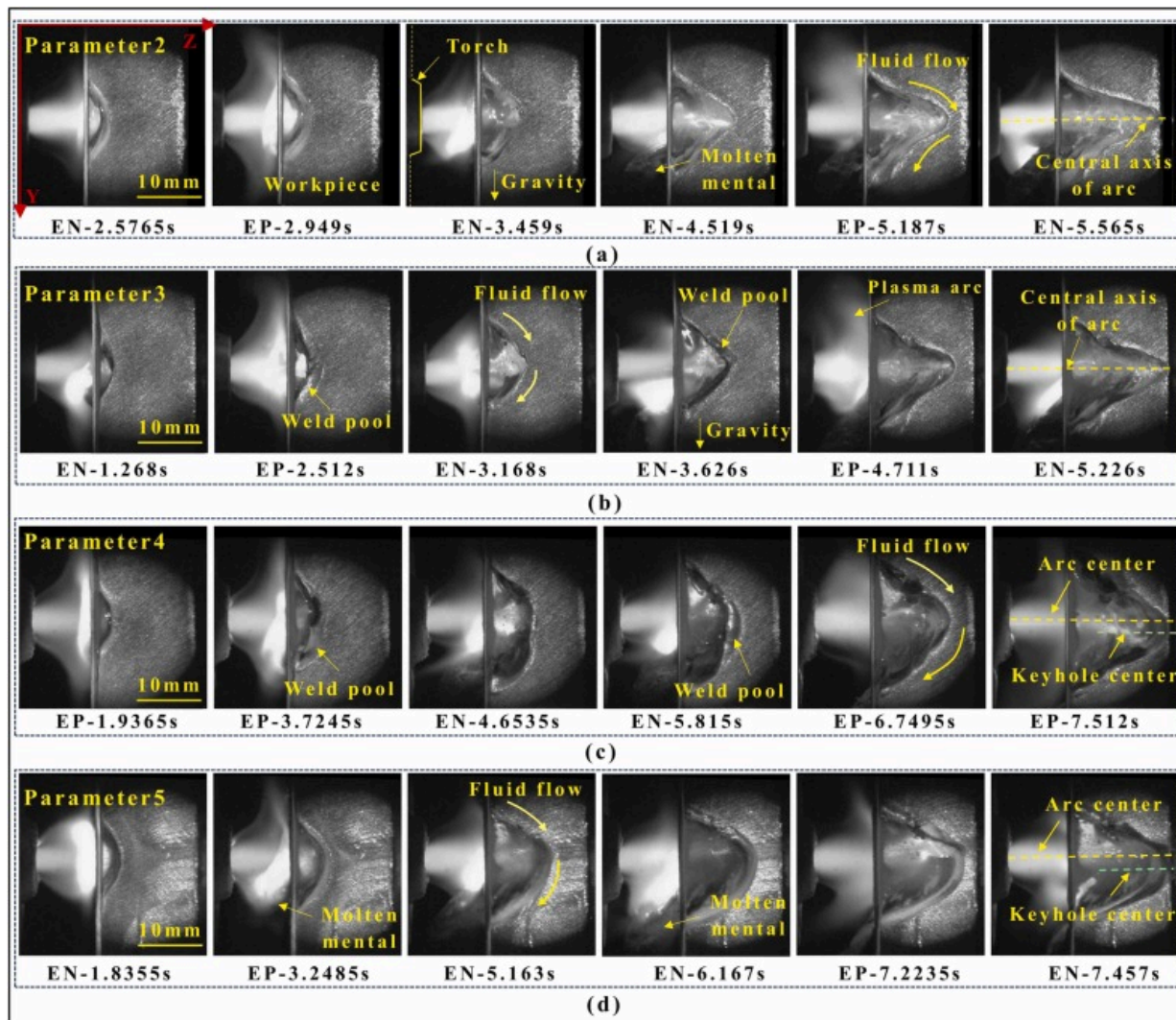
As the perforation depth increases, arc energy is gradually redirected to melt the sidewalls of the keyhole, transitioning the process into the second quasi-steady stage (Q2). Once the sidewalls are sufficiently melted, the arc energy becomes concentrated again at the bottom of the keyhole, triggering the third rapid growth stage (R3). However, due to the extended perforation depth and increased thermal dissipation, the rate of depth increase in R3 is notably slower compared to that in R2. After the digging depth

reaches 11.48 mm, an energy accumulation phase lasting approximately 0.22 s is observed (Q3), followed by the onset of the BP phenomenon. Similar BP event has been reported by Chen et al. [10] in VPPAW of a 5 mm-thick aluminum workpiece, where the phenomenon is attributed to localized energy accumulation that significantly increased the velocity of the molten metal. A comparable mechanism is considered responsible for the BP event observed in this study. Additionally, during the BP stage, the WPTBK becomes extremely thin, permitting direct interaction between the high-energy arc and the underlying solid aluminum alloy, as shown in Fig. 6. Consequently, the dominant heat transfer mode transitions from conduction to arc-driven convection. Wu et al. also demonstrated the presence of two convection currents in the weld pool during the digging stage [51]. This direct arc-metal contact facilitates rapid melting and contributes to the occurrence of the BP. In this study, the penetration depth during the BP stage reaches 4.37 mm, accounting for approximately 25.5 % of the full workpiece thickness. This observation further supports the energy accumulation hypothesis, as the depth of the BP remains confined within a relatively limited range.

3.4. Adjusting welding parameters in variable polarity plasma arc welding

To systematically investigate the evolution of the keyhole and weld pool during the digging stage under varying thermal and mechanical conditions, a series of welding parameters were developed based on the standard parameter. Fig. 15 presents the keyhole and weld pool formation processes corresponding to representative parameter configurations. The Y- and Z- directional motions are consistent with the standard parameter. Under parameter 2, an increased plasma gas flow rate results in an enhanced arc force, exerting a significant influence on keyhole penetration dynamics. As illustrated in Fig. 15a, the total penetration time is 5.52 s, which is 0.837 s shorter than that observed under standard parameter. During the initial stage (prior to 2.949 s), molten metal accumulation is evident. Thereafter, the weld pool boundary and the keyhole boundary at the bottom of the keyhole become nearly coincident, rendering the WPTBK nearly indistinguishable. Additionally, the keyhole cross-sectional area is markedly smaller than that observed under the standard parameter, a consequence of the intensified arc force,

which substantially improved penetration efficiency. Under parameter 3, an increase in welding current further reduces the penetration time to 5.23 s, which is even 0.29 s shorter than that under parameter 2. The elevated current enhances arc energy density, thereby offering a more effective means of improving penetration efficiency. As shown in Fig. 15b, a noticeable deviation in keyhole morphology emerges during the later stage of the digging process, after 4.711 s. The keyhole geometry along the Z-axis exhibits a distinct dual-region structure, forming a characteristic ‘trumpet-shaped’ configuration, with the left section (closer to the arc entrance) appearing broader and the right section (closer to the exit) narrower. Even upon full penetration, the keyhole demonstrates near-symmetry about the central axis of the arc. These morphological features directly reflect the intensified thermal action induced by the higher arc energy, highlighting its critical role in penetration characteristics.



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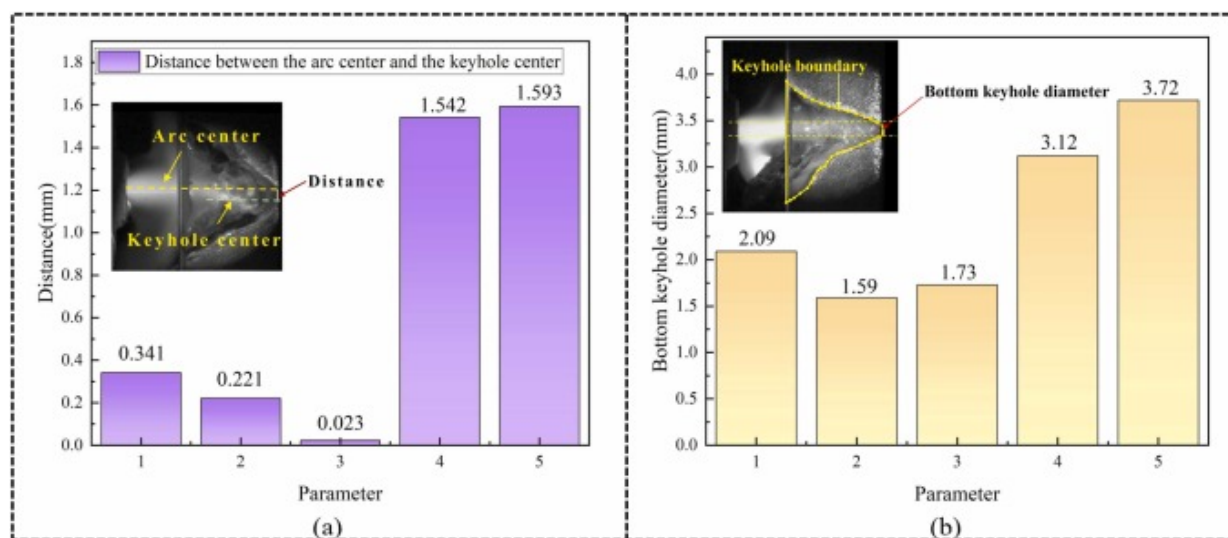
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Fig. 15. Keyhole and weld pool formation processes under representative parameters: (a) Parameter 2, (b) Parameter 3, (c) Parameter 4, (d) Parameter 5.

Conversely, when the welding current is decreased, as under parameter 4, the digging time is extended to 7.514 s, 1.157 s longer compared to the standard parameter. As shown

in Fig. 15c, molten metal accumulation is evident at 3.7245 s and 5.815 s, primarily attributing to the reduced heat input associated with the lower current, which decreases the fluidity of the molten metal and promotes its accumulation within the keyhole. Interestingly, the cross-sectional area of the keyhole exceeds that observed with the standard parameter. This is due to the extended digging duration, which allows the arc to act within the keyhole for a longer period, enabling the accumulated thermal energy to further promote melting solid metal along the Y-axis. Similarly, under parameter 5, characterized by a reduced plasma gas flow rate, the total penetration time increases to 7.42 s, as illustrated in Fig. 15d. A distinctive feature under this condition is the significant enlargement of the keyhole diameter at the bottom region upon full penetration. These phenomena indicate a reduced arc penetration capability under these two parameters, resulting in weaker penetration performance.

The geometric dimensions of the keyhole serve as a direct reflection of the dynamic behaviors of both the keyhole and the weld pool. In contrast to medium-thickness plates in VPPAW, a notable feature observed during thick aluminum alloy is the spatial misalignment between the arc center and the bottom keyhole center. Fig. 16a quantitatively presents the offset distance under different welding parameters. The measured offset distances for parameter 1 through 5 are 0.341 mm, 0.221 mm, 0.023 mm, 1.542 mm, and 1.593 mm, respectively. The misalignment is primarily attributed to the extended digging distance inherent to thick plates, which induces a more complex and asymmetric energy transfer process. An increase in welding current or plasma gas flow rate enhances the thermal and mechanical effects exerted by the arc on the workpiece, thereby reducing the offset distance, as observed in parameters 2 and 3. On the other hand, a reduction in these parameters diminishes the influence of arc, resulting in a pronounced increase in offset distance, as demonstrated in parameters 4 and 5.



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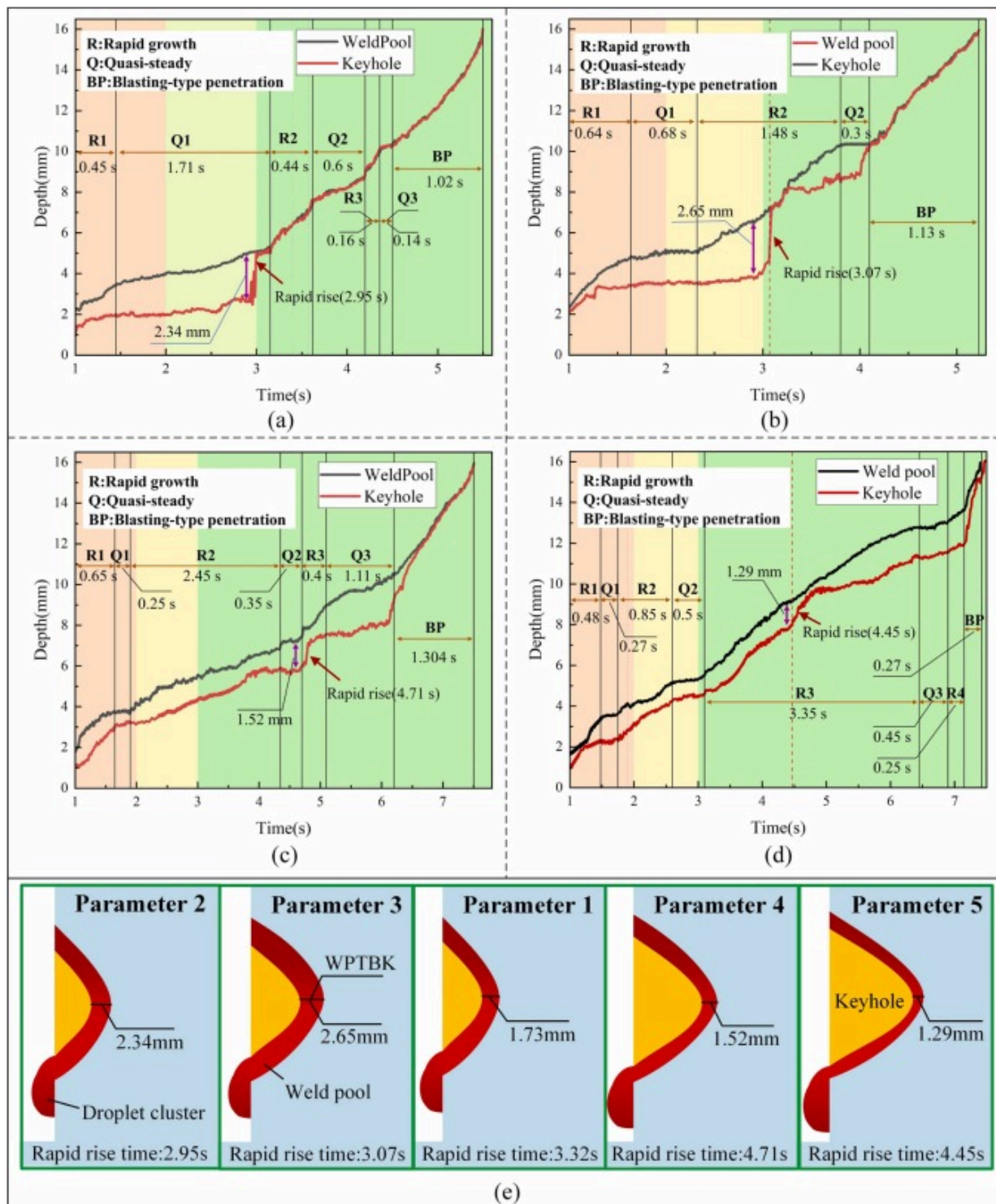
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Fig. 16. Geometric dimensions of the keyhole and plasma arc under different parameters. (a) Distance between the keyhole center and arc center, (b) Bottom keyhole diameter.

Fig. 16b displays the diameters of the bottom keyhole at the moment of arc breakthrough for each parameter, measured as 2.09 mm, 1.59 mm, 1.73 mm, 3.12 mm, and 3.72 mm, respectively. Interestingly, this trend deviates from typical observations in VPPAW of medium-thick aluminum alloys, where increasing welding current or ionized gas flow generally leads to an enlarged bottom keyhole due to enhanced arc penetration capability. The deviation observed in VPPAW for thick aluminum alloy is attributed to the extended digging distance. In such cases, reduced current or plasma gas flow prolongs the digging time, allowing the high-energy arc to dwell within the keyhole for a longer period. The sustained energy input results in greater thermal accumulation, which subsequently leads to an enlargement of the bottom keyhole.

Fig. 17 presents the keyhole and weld pool formation characteristics under representative parameters, focusing on the differences from the standard parameter and the distinctive

characteristics observed for each parameter. Fig. 17a illustrates the scenario with an increased plasma gas flow rate, where the oscillatory behavior and BP event persist. An increased plasma gas flow rate results in elevated arc pressure, thereby inducing a maximum weld pool thickness of 2.34 mm, which is 0.61 mm greater than that observed under the standard parameter. Moreover, the rapid rise in keyhole depth is initiated earlier, occurring at 2.95 s, approximately 0.37 s earlier than under the standard condition. Unlike the standard parameter, once the WPTBK reaches its maximum value, the keyhole depth does not undergo a gradual increase, but instead experiences a rapid rise. This phenomenon arises from the enlarged molten metal volume, thereby intensifying the impact of gravity on the weld pool dynamics. As a result, the molten metal flows out rapidly, accelerating the detachment of the external droplet cluster, which in turn further contributes to the rapid increase in keyhole depth. Additionally, keyhole depth fluctuations are observed during the rapid growth phase of the keyhole depth. These oscillations arise from the enhanced arc force, which induces a backflow of molten metal from the upper portion of the keyhole towards the bottom, thereby generating oscillations in the keyhole depth. Interestingly, beyond 3.0 s, the keyhole boundary increasingly overlaps with the weld pool boundary, transitioning to a convective heat transfer mode, thereby significantly improving the digging efficiency.



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Fig. 17. Keyhole and weld pool formation characteristics under representative parameters. (a-d) Quantitative analysis of keyhole and weld pool depths under parameter 2–5. (e) Schematic illustration of features on thermal equilibrium state.

[Fig. 17b](#) quantitatively illustrates the evolution of keyhole and weld pool depths under parameter 3, which corresponds to an elevated welding current. The weld pool reaches the initial heat and force equilibrium at 1.64 s. After an energy accumulation period of 0.68 s, the high-energy arc induced by the elevated welding current causes the weld pool depth to continue increasing. At 3.07 s, 0.25 s earlier than the standard parameter, the force equilibrium is disrupted, resulting in the rapid flow of molten metal. The WPTBK reaches its maximum thickness of 2.65 mm, which is 0.92 mm thicker than that observed under standard parameter. Similar to parameter 2, the rapid outflow of a large volume of molten metal following the disruption of force equilibrium causes the weld pool depth to overlap with the keyhole depth. This morphological transition is accompanied by a shift in the dominant heat transfer mechanism from conduction to convection, resulting in a substantial increase in the growth rate of weld pool depth. As the weld pool continues to thicken, the heat transfer mode shifts back to conduction, causing a gradual reduction in the growth rate of weld pool depth. The oscillatory behavior exhibits only two alternating cycles of rapid growth and quasi-steady states. The elevated welding current enhances the arc energy input, thereby improving the digging efficiency and streamlining the keyhole formation process.

[Fig. 17c](#) presents the scenario in which a reduction in welding current leads to a diminished arc energy and force effect relative to the standard parameter. Under this condition, force equilibrium is disrupted when the WPTBK reaches 1.52 mm, 0.26 mm thinner than that of the standard case. Once this critical thickness is attained, the keyhole depth undergoes a rapid rise. The onset of force imbalance is delayed to 4.71 s,

representing a 1.39 s lag relative to the standard parameter. Due to the prolonged accumulation of molten metal outflow prior to the disruption, a sizable droplet cluster has already formed. Upon loss of equilibrium, the rapid outflow of molten metal accelerates the detachment of the external droplet cluster, corresponding to the observed rapid rise in keyhole depth. Additionally, a prolonged energy accumulation phase (Q3) of 1.61 s is observed before the BP event, due to the need for more energy accumulation after the reduction in arc energy.

Fig. 17d corresponds to the scenario with a reduced plasma gas flow rate. The rapid rise occurs at 4.45 s, delayed by 1.13 s compared to the standard parameter. Notably, the duration of the rapid rise in keyhole depth is significantly longer than those observed under other conditions. This behavior is primarily attributed to the limited WPTBK prior to the disruption of force equilibrium, which measures only 1.29 mm and is 0.44 mm thinner than that in the standard case. Additionally, as shown in Fig. 15d, the rear wall of the keyhole forms a smaller angle with respect to the horizontal plane, which elongates the flow path and reduces the velocity of molten metal migrating from the keyhole interior to the external droplet cluster. As a result, the droplet cluster exhibits reduced responsiveness to changes in keyhole depth. A distinct transitional stage, denoted as R4, is also identified preceding the onset of the BP event. Furthermore, the final BP depth is substantially reduced, reaching only 2.8 mm, indicating decreased penetration efficiency under this condition.

The phenomenon of rapid keyhole depth increase, consistently associated with the detachment of droplet cluster from the workpiece surface, is observed across a range of experimental parameters. Oscillatory behavior, followed by the occurrence of a BP event, is identified under various welding conditions, primarily driven by the extended digging distance in thick workpieces. These observations validate the proposed mechanism governing the dynamic behaviors of keyhole and weld pool. Parameter-specific analysis further reveals that an increase in arc force or energy corresponds to a rise in WPTBK, which in turn accelerates the detachment of the droplet cluster. Conversely, a reduction in arc force or energy results in a thinner WPTBK, causing a delayed detachment of the

droplet cluster. Fig. 17e presents a comparative analysis of the WPTBK thickness at the thermal equilibrium state and rapid rise time of keyhole depth under various welding parameters. Correspondingly, the timing of droplet cluster detachment, which varies with different welding conditions, proves to be a valuable indicator for evaluating the appropriateness of welding parameter. This understanding provides a crucial basis for the development of intelligent welding control strategies, enabling real-time optimization of process parameter to improve welding stability.

4. Conclusions and future work

This study proposed a novel sandwich structure methodology to visualize and quantify the internal evolution of the keyhole and weld pool during the digging stage in variable polarity plasma arc welding (VPPAW) of thick aluminum alloy. It clearly reveals the internal dynamic behaviors of the keyhole and weld pool under different thermal-force conditions, providing a key guide for the optimization of deep penetration keyhole welding. The key conclusions are as follows:

- (1) The weld pool depth during digging exhibits a distinct oscillatory evolution consisting of three alternating cycles of rapid growth and quasi-steady behavior, ultimately transitioning into a blasting-type penetration. The quasi-steady behavior is attributed to the increased heat dissipation induced by the thickened weld pool thickness at the bottom of the keyhole (WPTBK) and the inherently high thermal conductivity of aluminum alloy, which collectively drive the system toward a temporary heat-force equilibrium. As the WPTBK thickens, the force equilibrium is disrupted, triggering a rapid outflow of molten metal from the keyhole. This disruption in force balance subsequently breaks the thermal-force equilibrium, enabling the plasma arc to reestablish effective downward penetration.
- (2) A distinct phenomenon observed during the digging stage in VPPAW of thick aluminum alloy is the formation of a molten droplet cluster in the lower

side of the keyhole. The internal behaviors of both keyhole depth rapidly rises and the blasting-type penetration can be effectively reflected through the external features of the surface weld pool and droplet cluster area, offering valuable information for intelligent control of the VPPAW.

- (3) Under varied welding parameters, oscillatory behavior and blasting-type penetration persist. An increase in arc force or energy leads to a rise in WPTBK and an earlier detachment of the droplet cluster, whereas a decrease in arc force or energy results in the opposite effect.
- (4) The outflow of molten metal from the keyhole interior is beneficial for the keyhole digging, while excessive expulsion of molten material results in welding defects such as undercuts and poor fusion. The failure to maintain an optimal balance of molten metal during the digging stage is a key factor contributing to the narrow process window in VPPAW of thick workpieces.

Building upon the identified evolution patterns of the keyhole and weld pool, as well as the established correlation between external characteristics and internal evolution of the keyhole and weld pool, future work will focus on developing intelligent control strategies for VPPAW of thick plates to enhance process stability.

CRedit authorship contribution statement

Wenlong Li: Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shujun Chen:** Supervision, Resources. **Di Yang:** Supervision, Investigation. **Jiankang Song:** Writing – review & editing, Methodology, Investigation. **Bin Xu:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Fan Jiang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Zhenzhen Zhang:** Software, Investigation. **Xinqiang Ma:** Supervision, Funding acquisition. **Wei Cheng:** Validation, Supervision, Funding acquisition. **Haowen Suo:** Writing – review & editing, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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